TRACE SPARSITY AND ARITHMETICITY FOR COMPACT HYPERBOLIC SURFACES

ABSTRACT. Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be a torsion-free cocompact lattice, and let $T(\Gamma)$ be the set of traces of the full inverse image of Γ in $\mathrm{SL}_2(\mathbb{R})$. We prove arithmeticity under the critical trace-growth exponent:

$$\limsup_{X\to\infty} \frac{\log \#\big(T(\Gamma)\cap[-X,X]\big)}{\log X} \le 1 \quad\Longrightarrow\quad \Gamma \text{ is arithmetic.}$$

Equivalently, the hypothesis is the family of bounds $\#(T(\Gamma)\cap[-X,X]) = O_{\Gamma,\eta}(X^{1+\eta})$ for every $\eta > 0$. In particular, linear growth and bounded clustering of the signed trace set force arithmeticity. These consequences are the compact trace formulations of the conjectures of Schmutz and Sarnak.

The main estimate compares translation length at the original Fuchsian character with arithmetic expansion at an algebraic specialization. Let x be the Fuchsian character of Γ, let y be an algebraic Fuchsian character in the rational Zariski closure of x, and let k_y be the trace field of the subgroup generated by squares. For every element b of that subgroup we prove

$$\mathcal{J}_{k_y}(\mathrm{tr}_y\, b) \le \frac{\ell_x(b)}{2},$$

where $\mathcal{J}_k$ is the sum of the local expansion exponents of the associated Chebyshev recurrence. When the original character is algebraic, the distinguished real summand of $\mathcal{J}_{k_x}(\mathrm{tr}_x\, b)$ already equals $\ell_x(b)/2$. Equality therefore holds, all other local exponents vanish, and Takeuchi's criterion gives arithmeticity.

To prove this estimate, we fix an index in the Chebyshev recurrence and vary the group element. Roth's theorem bounds from below the reduced divisor of two consecutive recurrence terms. Expansion in congruence quotients modulo p^2 allows us to choose elements for which no prime outside a fixed finite set occurs in that divisor to an unbounded power. We obtain the corresponding upper bound by combining three counts: a uniform bound for fibers of two trace functions, a congruence count for moduli with bounded prime exponents, and a bound on the fibers of a map from length-two paths in a bipartite trace graph. Algebraic specialization and Thurston's asymmetric length-spectrum metric then show that the original Fuchsian character cannot be transcendental. Section 4 states the analytic and arithmetic results from the literature, including all uniformity needed in these counts.

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Date: Working paper, 8 October 2026.

This working paper was developed through discussions and model prompting by Alexander Kolpakov (UATX) and Nikolay Bogachev (University of Toronto).

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1. Introduction

Let S be a closed orientable hyperbolic surface and let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be the corresponding torsion-free cocompact lattice. Write $\widetilde{\Gamma}$ for its full inverse image in $\mathrm{SL}_2(\mathbb{R})$, and define the *signed trace set*

$$T(\Gamma) = \{\operatorname{tr} g : g \in \widetilde{\Gamma}\}.$$

Thus $T(\Gamma) = -T(\Gamma)$; it is a set of values, and repeated traces are not counted with multiplicity. Put

$$N_\Gamma(X) = \#(T(\Gamma) \cap [-X, X]) \quad (X \geq 2)$$

and define its *upper trace-growth exponent* by

$$\overline{\delta}_T(\Gamma) = \limsup_{X\to\infty} \frac{\log N_\Gamma(X)}{\log X}. \tag{1}$$

We say that the trace set satisfies the *critical trace-growth bound* if $\overline{\delta}_T(\Gamma) \leq 1$. This is equivalent to

$$N_\Gamma(X) = O_{\Gamma,\eta}(X^{1+\eta}) \qquad \text{for every } \eta > 0, \tag{2}$$

or, more briefly, to the upper bound $N_\Gamma(X) \leq X^{1+o(1)}$. We say that $T(\Gamma)$ has *linear growth* if

$$\#(T(\Gamma) \cap [-X, X]) = O_\Gamma(X) \qquad (X \longrightarrow \infty). \tag{3}$$

It has *bounded clustering* if

$$\sup_{u\in\mathbb{R}} \#(T(\Gamma) \cap [u-1, u+1]) < \infty. \tag{4}$$

Linear growth implies the critical trace-growth bound. The choice of radius in (4) is immaterial: an interval of one fixed positive radius can be covered by finitely many intervals of any other fixed positive radius. Covering $[-X, X]$ by $O(X)$ intervals of radius one then shows that bounded clustering implies linear growth.

If $g \in \widetilde{\Gamma}$ projects to a nonidentity element, its translation length satisfies

$$|\operatorname{tr} g| = 2\cosh\frac{\ell(g)}{2}. \tag{5}$$

Consequently, if $L_\Gamma(R)$ counts distinct positive translation lengths at most R, including lengths of iterates, then

$$N_\Gamma(X) = 2 + 2L_\Gamma\big(2\operatorname{arcosh}(X/2)\big).$$

The constant term counts the central traces ± 2, and the factor two counts the two signs of each hyperbolic trace. Since $2\operatorname{arcosh}(X/2) = 2\log X + O(1)$, this converts the trace statements used here into the usual length formulations in [21, 22].

For context, Luo and Sarnak proved that arithmetic cofinite Fuchsian groups have bounded clustering [15]. Geninska and Leuzinger proved the converse under bounded clustering for nonuniform lattices [6], and Hao subsequently proved the stronger linear-growth converse in the nonuniform case [9]. For closed surfaces of genus at least six, Hao also proved a superlinear generic lower bound outside a countable union of positive-codimension algebraic subsets of Teichmüller space [10]. The compact case for all hyperbolic metrics addressed here was not covered by those results.

We recall the arithmetic terminology used in the conclusion. A quaternion algebra B over a number field k is a four-dimensional central simple k-algebra. It has a reduced norm $\mathrm{Nrd} : B \to k$, which becomes the determinant after any field extension that identifies B with M_2. In the Fuchsian case, B/k is called *admissible* when k is totally real, B is split at one distinguished real place, and B is ramified at every other real place. An *order* $\mathcal{O} \subset B$ is a subring containing 1 which is also a full $\mathcal{O}_k$-lattice. Its norm-one group is

$$\mathcal{O}^1 = \{u \in \mathcal{O} : \mathrm{Nrd}(u) = 1\}.$$

Using the distinguished splitting $B \otimes_{k,v_0} \mathbb{R} \simeq M_2(\mathbb{R})$, the projective image of $\mathcal{O}^1$ is a lattice in $\mathrm{PSL}_2(\mathbb{R})$. A Fuchsian group is *arithmetic* if it is commensurable with such a projective norm-one group. Two groups are commensurable when their intersection has finite index in each. A group is *derived from a quaternion algebra* when, after conjugation, it is contained with finite index in the projective norm-one group of an order. These formulations define the same commensurability class.

Our main theorem is the following.

Theorem 1.1 (trace sparsity forces arithmeticity)**.** *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be torsion-free and cocompact. If its signed trace set satisfies $\overline{\delta}_T(\Gamma) \le 1$, then Γ is arithmetic.*

Corollary 1.2 (compact Schmutz and Sarnak conjectures and an orbifold extension)**.** *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be any cocompact Fuchsian lattice, possibly with torsion. The critical trace-growth bound forces Γ to be arithmetic. In particular, linear growth or bounded clustering of the signed trace set forces arithmeticity. Thus the two compact-surface conjectures hold, and the same conclusions extend to compact orientable hyperbolic orbifolds.*

Theorem 1.1 is the torsion-free part of Corollary 1.2. Selberg's lemma and invariance of arithmeticity under commensurability give the extension to lattices with torsion. Linear growth implies the critical bound, while the covering argument following (4) shows that bounded clustering implies linear growth. The main step is a local inequality which bounds all arithmetic expansion rates at an algebraic Fuchsian character by the geometric length at the original character.

Let $\Pi = \pi_1(S)$, and let

$$\Pi^{(2)} = \langle \gamma^2 : \gamma \in \Pi \rangle.$$

The quotient $\Pi/\Pi^{(2)}$ is the finite elementary abelian group $H_1(S, \mathbb{F}_2)$, so $\Pi^{(2)}$ has finite index. A lift of a PSL_2-representation to SL_2 is not unique, but two lifts differ by a character $\Pi \to \{\pm 1\}$. Such a character is trivial on $\Pi^{(2)}$. Hence traces on $\Pi^{(2)}$, and the field generated by them, do not depend on the chosen lift. For a marked Fuchsian group Γ, we similarly write

$$\Gamma^{(2)} = \langle g^2 : g \in \Gamma \rangle;$$

it is the projective image of $\Pi^{(2)}$. If $\widetilde{\Gamma} < \mathrm{SL}_2(\mathbb{R})$ is the full inverse image, then

$$k_\Gamma = \mathbb{Q}\big((\mathrm{tr}\, \widetilde{g})^2 : \widetilde{g} \in \widetilde{\Gamma}\big)$$

denotes the invariant trace field.

For a number field k and $t \in k$, Section 3 defines a nonnegative quantity $\mathcal{J}_k(t)$. When a distinguished real embedding sends t to the trace of a hyperbolic element of translation length L, that one embedding contributes exactly $L/2$ to $\mathcal{J}_k(t)$.

Theorem 1.3 (local expansion for algebraic points in the rational closure)**.** *Let x be a faithful cocompact Fuchsian character of the closed surface group Π, and assume that its signed trace set satisfies the critical trace-growth bound. Let $\mathcal{X}$ be the affine $\mathbb{Q}$-variety of liftable PSL_2-characters described in Section 2, and let $Z_x \subset \mathcal{X}$ be the rational Zariski closure of x. Let $y \in Z_x(\overline{\mathbb{Q}})$ be an algebraic faithful cocompact Fuchsian character, and put*

$$k_y = \mathbb{Q}(\mathrm{tr}_y\, h : h \in \Pi^{(2)}).$$

Then, for every $b \in \Pi^{(2)}$,

$$\mathcal{J}_{k_y}(\mathrm{tr}_y\, b) \le \frac{\ell_x(b)}{2}. \tag{6}$$

For the identity element both sides are zero.

Corollary 1.4 (rigidity among algebraic Fuchsian specializations)**.** *Under the hypotheses of Theorem 1.3, assume in addition that y lies in the same Fuchsian component as x. This holds, in particular, for the nearby specializations supplied by Proposition 2.3. Then the two marked Fuchsian characters are equal: $y = x$.*

Proof. For every $\gamma \in \Pi$, apply Theorem 1.3 to $b = \gamma^2 \in \Pi^{(2)}$. The distinguished real place of k_y contributes $\ell_y(b)/2$ to the nonnegative sum $\mathcal{J}_{k_y}(\mathrm{tr}_y\, b)$. Hence

$$\ell_y(\gamma) = \frac{\ell_y(\gamma^2)}{2} \le \mathcal{J}_{k_y}(\mathrm{tr}_y(\gamma^2)) \le \frac{\ell_x(\gamma^2)}{2} = \ell_x(\gamma).$$

Thurston's minimal-stretch theorem identifies the logarithm of the least Lipschitz constant in the marked homotopy class with

$$d_{\mathrm{Th}}(x, y) = \log \sup_{\gamma \ne 1} \frac{\ell_y(\gamma)}{\ell_x(\gamma)}.$$

It defines a separating asymmetric metric on Teichmüller space [24, Proposition 2.1 and Theorem 8.5]. The displayed domination therefore gives $d_{\mathrm{Th}}(x, y) \le 0$. Since this metric is nonnegative, it is zero, and its separation property gives $x = y$. □

Here and below an element of $\Pi^{(2)}$ and its lift to SL_2 are denoted by the same letter when the choice of sign is irrelevant. Since $\mathcal{J}_k(-t) = \mathcal{J}_k(t)$, inequality (6) is independent of that sign.

We next fix notation used in the proof. For a number field k and a finite set S of its places, let

$$\mathcal{O}_{k,S} = \{a \in k : |a|_v \le 1 \text{ for every finite place } v \notin S\} \tag{7}$$

be the ring of S-integers. Thus archimedean places may be included in S, but do not affect the ring. A nonzero integral ideal of $\mathcal{O}_{k,S}$ has a unique factorization $\mathfrak{a} = \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{e_\mathfrak{p}}$, and its absolute norm is

$$\mathrm{N}\,\mathfrak{a} = \#(\mathcal{O}_{k,S}/\mathfrak{a}) = \prod_{\mathfrak{p} \notin S} \#(\mathcal{O}_k/\mathfrak{p})^{e_\mathfrak{p}}. \tag{8}$$

In particular, places in S do not contribute to the norm of a principal S-ideal.

Choose a basepoint $o \in \mathbb{H}^2$. The notation $d_x(o, go)$ means hyperbolic displacement under the representation with character x, and $\ell_x(g) = \inf_z d_x(z, gz)$ is its translation length. Word length, denoted by $|g|$, is always taken with respect to a finite symmetric generating set fixed before the relevant argument. In the main construction we use the image under ρ_y of one fixed generating set of $\Pi^{(2)}$. Replacing this generating set changes only constants, because any two word metrics on a finitely generated group are quasi-isometric.

Subscripts on asymptotic notation specify the data on which the constants may depend. For example, $r_m = o_\mathcal{D}(m)$ means $r_m/m \to 0$ while the data $\mathcal{D}$ remain fixed, and $O_\mathcal{D}(1)$ denotes a quantity bounded in terms of those data. An unsubscripted O- or o-term inside a proof is uniform after all objects fixed earlier in that proof have been fixed. Whenever auxiliary data vary in a later limiting operation, we state the order of limits explicitly.

Here is the argument for Theorem 1.3 in outline. For

$$q_0(T) = 0, \qquad q_1(T) = 1, \qquad q_{n+1}(T) = Tq_n(T) - q_{n-1}(T),$$

fix $N \geq 3$ and put

$$F_N(T) = q_{N-1}(T)q_N(T).$$

Starting from b, we choose $a_m = b^m h_m$, where h_m has word length at most θm in the fixed generating set. If $s_m = \mathrm{tr}_y(a_m)$, this choice has three properties:

$$\mathcal{J}_{k_y}(s_m) \geq m\mathcal{J}_{k_y}(\mathrm{tr}_y\, b) - O(\theta m) - O(1),$$
$$d_x(o, a_m o) \leq m\ell_x(b) + O(\theta m) + O(1),$$
$$F_N(s_m)\mathcal{O}_{k_y,S} \text{ has uniformly bounded prime exponents.}$$

Roth's theorem then gives the lower estimate

$$\log \mathrm{N}\big(F_N(s_m)\mathcal{O}_{k_y,S}\big) \geq (2N - 4 - \epsilon)\mathcal{J}_{k_y}(s_m) - O(1). \tag{9}$$

The trace-graph and congruence argument gives the upper estimate

$$\log \mathrm{N}\big(F_N(s_m)\mathcal{O}_{k_y,S}\big) \leq \frac{2N-1}{2} d_x(o, a_m o) + o(m). \tag{10}$$

After division by m, we let $m \to \infty$, then $\theta, \epsilon \to 0$, and finally $N \to \infty$. Since

$$\frac{(2N-1)/2}{2N - 4 - \epsilon} \longrightarrow \frac{1}{2},$$

this proves (6).

The limits must be taken in this order. The finite set of excluded places, the exponent bound in the sieve, and the ball radius used for congruence counting may depend on $N, \theta, \epsilon, b, y$, but they are fixed before m tends to infinity.

The following sections prove these estimates. Section 4 states the results from the literature in the forms used here. The superapproximation needed to select a_m concerns only squares of primes; the later congruence count uses composite moduli whose prime exponents are bounded by one fixed constant. No expansion theorem for arbitrary prime powers or arbitrary ideals is assumed.

2. Fuchsian characters and rational closure

2.1. **Characters, lifts, and trace functions.** Fix a presentation of the closed surface group Π. Let $\mathcal{X}$ be the affine $\mathbb{Q}$-variety of liftable PSL_2-characters obtained from the SL_2-character variety by quotienting by the finite group of sign twists $\mathrm{Hom}(\Pi, \{\pm 1\})$. Thus $\mathbb{Q}[\mathcal{X}]$ is the invariant subring of the SL_2-character algebra. It is generated by the adjoint characters, or equivalently by the functions $\tau_\gamma = \mathrm{tr}^2(\gamma)$. Finite generation of matrix invariants and the trace algebra permits a finite collection of squared-trace generators [19, 12]. For the corresponding description of the PSL_2-character algebra, see [11, Proposition 2.2 and Theorem 1.3]. Those results are stated over $\mathbb{C}$: Proposition 2.2 gives generation by the functions τ_γ, and Theorem 1.3 identifies the closed points with squared-trace characters. Faithfully flat descent along $\mathbb{C}/\mathbb{Q}$ gives the same generation statement for the chosen $\mathbb{Q}$-model: a quotient of finitely generated $\mathbb{Q}$-algebras is an isomorphism if it becomes an isomorphism after tensoring with $\mathbb{C}$. Since the coordinate ring is finitely generated, finitely many words suffice. After an orientation has been fixed, the real Fuchsian component is naturally identified with Teichmüller space. Goldman's description by the extremal Euler-class components shows that this is a connected component of the real representation space; after quotienting by conjugation, it is therefore open in the smooth real locus of $\mathcal{X}$ [7, Corollary C and Section 10.4]. For $\gamma \in \Pi$, the function

$$\tau_\gamma(z) = \mathrm{tr}_z(\gamma)^2$$

is a well-defined regular function on $\mathcal{X}$, independent of a choice of lift. If $h \in \Pi^{(2)}$, then $\mathrm{tr}_z(h)$ itself is independent of the lift and is regular on $\mathcal{X}$. Alternatively, one may work on a fixed finite cover that parameterizes SL_2-lifts. Since the degree of this cover is fixed, passing to it changes each cardinality estimate below by at most a fixed factor. The finite generation statement we need is the following.

Lemma 2.1 (finite squared-trace coordinates and the invariant trace field)**.** *There are words $\gamma_1, \dots, \gamma_r \in \Pi$ such that the functions τ_{γ_i} generate $\mathbb{Q}[\mathcal{X}]$ as a $\mathbb{Q}$-algebra. For every nonelementary character y,*

$$\begin{aligned} k_y &= \mathbb{Q}(\mathrm{tr}_y\, h : h \in \Pi^{(2)}) \\ &= \mathbb{Q}(\mathrm{tr}_y(\gamma)^2 : \gamma \in \Pi) = \mathbb{Q}(\tau_{\gamma_1}(y), \dots, \tau_{\gamma_r}(y)). \end{aligned} \tag{11}$$

In particular, if y is an algebraic point with residue field $\mathbb{Q}(y)$, then

$$k_y \subseteq \mathbb{Q}(y), \qquad [k_y : \mathbb{Q}] \le [\mathbb{Q}(y) : \mathbb{Q}]. \tag{12}$$

Proof. By the choice of the affine model, its coordinate ring is generated by squared-trace functions. Since that ring is finitely generated, finitely many of them suffice; call them τ_{γ_i}. The first equality in (11) is the definition of the full invariant trace field. The second is the invariant-trace-field identity for a nonelementary subgroup of PSL_2; see [16, Lemma 3.5.6]. One inclusion is also visible from

$$\mathrm{tr}_y(\gamma^2) = \mathrm{tr}_y(\gamma)^2 - 2, \qquad \gamma^2 \in \Pi^{(2)}.$$

The finite set τ_{γ_i} generates every squared-trace function as an element of the coordinate ring, which gives the last equality. Evaluation of regular functions at an algebraic point takes values in its residue field, proving (12). □

2.2. **Rational closure and persistence of trace collisions.** Let $A = \mathbb{Q}[\mathcal{X}]$. For $x \in \mathcal{X}(\mathbb{R})$, define

$$I(x) = \{f \in A : f(x) = 0\}, \qquad Z_x = V(I(x)) \subset \mathcal{X}. \tag{13}$$

We call Z_x the *rational Zariski closure* of x. The terminology refers to the fact that it is the smallest closed subvariety defined over $\mathbb{Q}$ that contains x.

Lemma 2.2 (trace equalities persist on the rational closure)**.** *If $y \in Z_x$ and $\tau_\alpha(x) = \tau_\beta(x)$, then $\tau_\alpha(y) = \tau_\beta(y)$. More generally, every polynomial relation over $\mathbb{Q}$ among trace functions that holds at x holds at every point of Z_x.*

Proof. The difference $\tau_\alpha - \tau_\beta$ belongs to $I(x)$, and every element of $I(x)$ vanishes on $V(I(x))$. The same argument applies to an arbitrary polynomial relation. □

The use of squared traces in this lemma is important. At a second point y, a signed equality may acquire either sign, but a fiber of squared traces splits into at most two signed fibers. Thus passing from squared traces at x to signed traces at y increases the number of values by at most a factor of two.

Proposition 2.3 (algebraic specialization of bounded degree)**.** *Let x be a Fuchsian point. If $\dim Z_x > 0$, there are a positive integer D and distinct algebraic Fuchsian points $y_j \in Z_x(\overline{\mathbb{Q}}) \cap \mathcal{X}(\mathbb{R})$, all in the same Fuchsian component as x, such that*

$$y_j \longrightarrow x, \qquad [\mathbb{Q}(y_j) : \mathbb{Q}] \le D. \tag{14}$$

For every finite set $E \subset \Pi$,

$$\#\{\tau_\gamma(y_j) : \gamma \in E\} \le \#\{\tau_\gamma(x) : \gamma \in E\}. \tag{15}$$

If the signed trace set at x satisfies the critical trace-growth bound and

$$\mathcal{B}_R^x = \{\gamma \in \Pi : d_x(o, \gamma o) \le R\},$$

then there is a function $\omega_x(R) = o_x(R)$, independent of j, such that

$$\#\{\mathrm{tr}_{y_j}\gamma : \gamma \in \mathcal{B}_R^x\} \le \exp(R/2 + \omega_x(R)). \tag{16}$$

The same assertion holds after replacing Π by any fixed finite-index subgroup.

Proof. The ideal $I(x)$ is the kernel of the evaluation homomorphism $A \to \mathbb{R}$, hence it is prime and Z_x is irreducible. Moreover, x does not lie on a proper closed subvariety of Z_x defined over $\mathbb{Q}$: an additional equation defining such a subvariety would vanish at x and would therefore already belong to $I(x)$. Since the ground field has characteristic zero, the singular locus is a proper $\mathbb{Q}$-closed subset. Thus x is a smooth point of Z_x.

Put $d = \dim Z_x$. Choose a rational linear Noether normalization

$$\pi : Z_x \longrightarrow \mathbb{A}^d$$

whose differential is invertible at x. Such a choice exists because the admissible linear projections form a nonempty Zariski-open set, and rational points are dense in that parameter space. After removing the branch and nonflat loci, π is finite and étale in a neighborhood of x. The real inverse-function theorem gives a local real analytic inverse branch through x. Rational points are dense in $\mathbb{A}^d(\mathbb{R})$, so one may choose distinct rational points approaching $\pi(x)$ within the image of that branch and lift them along the branch. Each lifted point lies in a fiber of the finite morphism over a rational point. It is therefore algebraic of degree at most $\deg\pi$, and the lifted points converge to x. Since the Fuchsian component containing x is open, the branch may be restricted so that every y_j is faithful, cocompact, and lies in the same Fuchsian component as x.

Lemma 2.2 says that two elements of E with the same squared trace at x still have the same squared trace at y_j. Thus the partition of E by squared trace at y_j is coarser than the corresponding partition at x, which proves (15). If $\gamma \in \mathcal{B}_R^x$, then

$$|\,\mathrm{tr}_x\gamma| \le 2\cosh(R/2).$$

The critical trace-growth bound therefore gives

$$\#\{\mathrm{tr}_x\gamma : \gamma \in \mathcal{B}_R^x\} \le \exp\bigl(R/2 + o_x(R)\bigr).$$

Choose one function $\omega_x(R) = o_x(R)$ that absorbs this error and the factor two introduced when signs are restored. The same estimate holds for squared traces. At y_j, collision persistence shows that no more squared-trace values occur, and each squared trace has at most two signed square roots. This proves (16) with a bound independent of j. Passing to a fixed finite-index subgroup only restricts the set being counted. □

If $\dim Z_x = 0$, then the residue field of the point is finite over $\mathbb{Q}$, so x is algebraic. Proposition 2.3 is needed only when x is transcendental.

2.3. **The trace field and its algebraic group.** Let y be an algebraic faithful Fuchsian character and let ρ_y be one fixed SL_2-lift. Let $\Delta = \rho_y(\Pi^{(2)}) < \mathrm{SL}_2(\mathbb{R})$, and write $\overline{\Delta} < \mathrm{PSL}_2(\mathbb{R})$ for its projective image. Put

$$k = k_y = \mathbb{Q}(\mathrm{tr}\,\Delta).$$

The two groups play different roles in the congruence argument. The linear group Δ carries the signed trace, the integral model, and the reduction maps. The projective group $\overline{\Delta}$ is the cocompact Fuchsian lattice whose finite-index subgroups define surface covers. A metric ball consists of abstract elements of $\Pi^{(2)}$: their displacement is measured through the projective representation at x, while their traces are evaluated through ρ_y. Because the projective representation at y is faithful, $\rho_y|_{\Pi^{(2)}}$ is injective, and we identify Δ with $\Pi^{(2)}$ only as an abstract group. Thus, for $A \in \Delta$, its

x-displacement and word length mean those of the unique abstract element $\rho_y^{-1}(A)$. Faithfulness of the projective representation, together with torsion-freeness, implies $\Delta \cap \{\pm I\} = \{I\}$. Passing instead to a full inverse image adds at most the central element $-I$ and changes all quotient and cardinality estimates by a factor at most two.

Lemma 2.4 (algebraic matrix realization and the associated quaternion algebra)**.** *After conjugation in* $\mathrm{SL}_2(\mathbb{C})$*, every matrix in* Δ *has algebraic entries. Moreover, the* k*-span of* Δ *is a four-dimensional central simple* k*-algebra.*

Proof. Every trace $\mathrm{tr}_y h$, for $h \in \Pi^{(2)}$, is the value at the algebraic point y of a regular function defined over $\mathbb{Q}$, and is therefore algebraic. Choose $A, C \in \Delta$ such that A is hyperbolic and C preserves neither eigenline of A. Such a pair exists because Δ is nonelementary. Conjugate A to

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}.$$

The eigenvalue λ is algebraic because it satisfies $X^2 - (\mathrm{tr}\, A)X + 1 = 0$. Write $C = \left(\begin{smallmatrix} p & q \\ r & s \end{smallmatrix}\right)$. The two algebraic numbers $\mathrm{tr}\, C$ and $\mathrm{tr}(AC)$ determine p, s. Both q and r are nonzero, and $qr = ps - 1$ is algebraic. Conjugation by a diagonal matrix, which leaves A fixed, may therefore normalize q to $\mathrm{sgn}(q)$: explicitly, use the real matrix $\mathrm{diag}(|q|^{-1/2}, |q|^{1/2})$. The other off-diagonal entry then becomes $\mathrm{sgn}(q)\, qr$, and is therefore algebraic. Thus this real conjugation makes both A and C algebraic while preserving the distinguished embedding into $M_2(\mathbb{R})$.

For $H = \left(\begin{smallmatrix} x & y \\ z & w \end{smallmatrix}\right) \in \Delta$, the traces of H and AH determine x, w. After those entries are known, the traces of CH and ACH give the two equations

$$qz + ry \in \overline{\mathbb{Q}}, \qquad \lambda qz + \lambda^{-1} ry \in \overline{\mathbb{Q}}.$$

Their determinant is $qr(\lambda^{-1} - \lambda) \neq 0$, so y, z are algebraic. This proves the first assertion.

The four matrices I, A, C, AC form a basis of $M_2(\mathbb{C})$: the two diagonal entries are separated by I, A, and the two off-diagonal entries are separated by C, AC. The bilinear trace pairing $(X, Y) \mapsto \mathrm{tr}(XY)$ is nondegenerate on $M_2(\mathbb{C})$. Every entry of the Gram matrix for this basis belongs to $k = \mathbb{Q}(\mathrm{tr}\, \Delta)$, because Cayley–Hamilton rewrites powers of A and C using their traces and all remaining products are group elements whose traces lie in k. Pairing an arbitrary element of Δ against the four basis elements and solving the resulting linear system therefore gives coefficients in k. Products of elements of Δ are again in Δ, so its k-span is closed under multiplication and has this four-element basis. After extension to $\mathbb{C}$ it is $M_2(\mathbb{C})$; it is consequently a four-dimensional central simple k-algebra, or equivalently a quaternion algebra over k. □

The k-subalgebra of $M_2(\mathbb{R})$ generated by Δ is a quaternion algebra B/k, because Δ is nonelementary. After enlarging a finite set S of finite places, there is a smooth affine group scheme $\mathcal{H}/\mathcal{O}_{k,S}$, with generic fiber $\mathrm{SL}_1(B)$, such that

$$\Delta \subset \mathcal{H}(\mathcal{O}_{k,S}). \tag{17}$$

Indeed, choose an $\mathcal{O}_k$-order, or merely a full $\mathcal{O}_k$-lattice, in B. Enlarge S so that the finitely many chosen generators of Δ lie in its $\mathcal{O}_{k,S}$-localization, and then remove the finitely many further places at which the resulting reduced-norm-one model is not smooth. We always enlarge S by all finite places over a rational prime whenever one place over that prime is included. Such an S will be called *rationally saturated*. We also include all places above the rational primes ramified in k. Thus every rational prime outside S is unramified in k. Norms of S-ideals are the absolute norms defined in (8); equivalently, only their prime-to-S factors contribute.

The restriction-of-scalars group associated with (17) is

$$\mathbf{G} = \mathrm{Res}_{k/\mathbb{Q}}\, \mathrm{SL}_1(B).$$

By definition, restriction of scalars is the $\mathbb{Q}$-group characterized by

$$\mathbf{G}(R) = \mathrm{SL}_1(B)(k \otimes_{\mathbb{Q}} R)$$

for every $\mathbb{Q}$-algebra R. After extension to $\overline{\mathbb{Q}}$, it becomes one copy of SL_2 for every embedding $k \hookrightarrow \overline{\mathbb{Q}}$.

Lemma 2.5 (Zariski closure after restriction of scalars)**.** *The connected $\mathbb{Q}$-Zariski closure of the restriction-of-scalars image of Δ is $\mathbf{G}$.*

Proof. The group Δ is nonelementary, hence Zariski dense in $\mathrm{SL}_1(B)$ over k. Over $\overline{\mathbb{Q}}$, the group $\mathbf{G}$ is a product of SL_2-factors indexed by the embeddings $\sigma : k \hookrightarrow \overline{\mathbb{Q}}$, and the connected closure of Δ projects onto every factor.

A connected subgroup that projects onto every factor of a product of simple groups of type A_1 is a product of diagonal subgroups. Here is the relevant consequence of Goursat's lemma. For any two factors, the image of the subgroup in their product is either the full product or the graph of an isomorphism between the factors. At the level of Lie algebras, this follows because the kernel of either projection is an ideal in $\mathfrak{sl}_2$, and hence is either zero or all of $\mathfrak{sl}_2$. Declare two factors equivalent when their two-factor image is such a graph. The subgroup is then the product, over the equivalence classes, of copies of SL_2 embedded diagonally in the factors belonging to that class. Suppose that one such diagonal subgroup identified the factors corresponding to two different embeddings $\sigma \neq \tau$. On a finite-index subgroup $\Delta_0 < \Delta$, the two representations would then differ by an algebraic automorphism of SL_2. Every such automorphism preserves trace, and consequently

$$\sigma(\operatorname{tr} h) = \tau(\operatorname{tr} h) \qquad (h \in \Delta_0).$$

The invariant trace field of a nonelementary Fuchsian group is unchanged on passage to finite index; see [16, Theorem 3.3.4]. Hence

$$k = \mathbb{Q}\big(\operatorname{tr}(h^2) : h \in \Delta_0\big).$$

The displayed equality of traces for $h \in \Delta_0$, applied also to h^2, makes σ and τ agree on this generating set of k, forcing $\sigma = \tau$. Thus every diagonal block is a singleton, which proves the assertion. □

The choice of the full trace field is essential here. Over a larger coefficient field, the Zariski closure could identify distinct restriction-of-scalars factors along a diagonal subgroup. Its congruence quotients would then have smaller dimension, invalidating the quotient-size estimate used below.

3. Local expansion and the Chebyshev recurrence

3.1. **The local expansion functional.** Let k be a number field, and write M_k for its set of archimedean and nonarchimedean places. Absolute values are normalized to extend the usual absolute values on $\mathbb{Q}$, and $n_v = [k_v : \mathbb{Q}_v]$ denotes the local degree. For $t \in k$ and a place v, choose a root z_v in an algebraic closure of k_v of

$$z + z^{-1} = t^2 - 2. \tag{18}$$

Define

$$\chi_v(t) = \frac{1}{2} \log \max\{|z_v|_v, |z_v|_v^{-1}\}, \qquad \mathcal{J}_k(t) = \sum_{v \in M_k} n_v \chi_v(t). \tag{19}$$

The two roots are reciprocal, so replacing z_v by z_v^{-1} does not change $\chi_v(t)$; the extension of the absolute value to the finite splitting field is unique up to conjugacy and gives the same value. The element t is integral at all but finitely many finite places. At every finite place where t is integral, both roots are units, so only finitely many nonarchimedean terms in $\mathcal{J}_k(t)$ are nonzero.

Our absolute logarithmic Weil height is normalized by

$$[k:\mathbb{Q}]h(t) = \sum_{v\in M_k} n_v \log^+ |t|_v, \qquad \log^+ r = \max\{0, \log r\}. \tag{20}$$

The product formula in the same normalization is $\sum_v n_v \log |a|_v = 0$ for $a \in k^\times$.

Lemma 3.1 (basic properties and change of field)**.** *Every local summand in* (19) *is nonnegative, and* $\mathcal{J}_k(-t) = \mathcal{J}_k(t)$*. If* L/k *is a finite extension and* $t \in k$*, then*

$$\mathcal{J}_L(t) = [L:k]\mathcal{J}_k(t). \tag{21}$$

Proof. The first two assertions follow directly from the definition and from the fact that (18) depends only on t^2. If w is a place of L above v, the normalized absolute value restricts to the one at v, so $\chi_w(t) = \chi_v(t)$. The local-degree formula is $\sum_{w|v} n_w = [L:k]n_v$. Summing over v proves (21). □

The definition can also be expressed in terms of eigenvalues. If $\lambda + \lambda^{-1} = t$, then $z = \lambda^2$ satisfies (18), and hence

$$\chi_v(t) = \log \max\{|\lambda|_v, |\lambda|_v^{-1}\}. \tag{22}$$

At a real Fuchsian place, if $|t| = 2\cosh(L/2)$, one may take $|\lambda| = e^{L/2}$, so

$$\chi_{v_0}(t) = \frac{L}{2}. \tag{23}$$

Lemma 3.2 (comparison with Weil height)**.** *For every number field* k *and every* $t \in k$*,*

$$|\mathcal{J}_k(t) - [k:\mathbb{Q}]h(t)| \le [k:\mathbb{Q}]\log 2. \tag{24}$$

At a nonarchimedean place the two local contributions agree exactly.

Proof. Choose λ as in (22) and put $M_v = \max\{|\lambda|_v, |\lambda|_v^{-1}\} \ge 1$. At a nonarchimedean place, the ultrametric inequality applied to $t = \lambda + \lambda^{-1}$ shows that $|t|_v = M_v$ when $M_v > 1$, whereas $|t|_v \le 1$ when $M_v = 1$. Thus $\log M_v = \log^+ |t|_v$.

At an archimedean place,

$$|t|_v \le |\lambda|_v + |\lambda|_v^{-1} \le 2M_v,$$

while the quadratic formula gives

$$M_v \le |t|_v + 1 \le 2\max\{1, |t|_v\}.$$

Therefore

$$\left|\log M_v - \log^+ |t|_v\right| \le \log 2.$$

Multiply by n_v, sum over all places, and use $\sum_{v|\infty} n_v = [k:\mathbb{Q}]$. □

In particular, if $\mathcal{J}_k(t) = 0$, then t is an algebraic integer and every archimedean conjugate of t belongs to $[-2, 2]$. More generally, if the entire sum $\mathcal{J}_k(t)$ comes from one real place, then every finite-place term vanishes, so t is an algebraic integer, and every other archimedean conjugate belongs to $[-2, 2]$.

3.2. **Recurrence identities.** Define $q_n \in \mathbb{Z}[T]$ by

$$q_0 = 0, \qquad q_1 = 1, \qquad q_{n+1} = Tq_n - q_{n-1}. \tag{25}$$

Thus $q_n(T) = U_{n-1}(T/2)$, where U_j is the Chebyshev polynomial of the second kind.

Lemma 3.3 (matrix recurrence and Cassini identity)**.** *Let $A, G \in \mathrm{SL}_2(K)$, where K is a field, and put $s = \operatorname{tr} A$. For every $n \geq 1$,*

$$A^n = q_n(s)A - q_{n-1}(s)I, \tag{26}$$

$$\operatorname{tr}(A^n G) = q_n(s)\operatorname{tr}(AG) - q_{n-1}(s)\operatorname{tr} G. \tag{27}$$

Moreover,

$$q_n(T)^2 - q_{n-1}(T)q_{n+1}(T) = 1. \tag{28}$$

Consequently two consecutive specialized values $q_{n-1}(s), q_n(s)$ generate the unit ideal in every ring containing s.

Proof. Cayley–Hamilton gives $A^2 - sA + I = 0$. Induction using (25) proves (26); multiplication by G and taking traces gives (27). The left side of (28) is independent of n by the recurrence and equals one when $n = 1$. The final assertion follows directly from (28). □

Two forms of (27) will be used repeatedly. Put $P = q_{N-1}(s)$, $Q = q_N(s)$. Then

$$Q\operatorname{tr}(AG) - P\operatorname{tr} G = \operatorname{tr}(A^N G), \tag{29}$$

$$Q\operatorname{tr} G - P\operatorname{tr}(AG) = \operatorname{tr}(A^{-(N-1)}G). \tag{30}$$

The second identity follows by applying Cayley–Hamilton backwards, or by multiplying (26) by A^{-N}.

Because $q_j \in \mathbb{Z}[T]$, every $q_j(t)$ is S-integral whenever $t \in \mathcal{O}_{k,S}$. In the principal argument the recurrence index N is fixed and the trace varies; no asymptotic assertion in the recurrence index is needed.

4. External results used in the proof

This section records the analytic and arithmetic theorems used below, with their required uniformity made explicit. Standard geometric or algebraic facts used only once are stated where they enter the proof. The deductions from these results are proved in the subsequent sections.

4.1. **Diophantine approximation and height finiteness.** We begin by recalling the local notation in Roth's theorem. If $P = (x_0 : x_1)$ and $Q = (y_0 : y_1)$ are points of $\mathbb{P}^1(\overline{k_v})$, their chordal distance is

$$\delta_v(P, Q) = \frac{|x_0y_1 - x_1y_0|_v}{\|(x_0, x_1)\|_v \|(y_0, y_1)\|_v},$$

where the maximum norm is used at a nonarchimedean place and a Euclidean norm at an archimedean place. A local projective proximity function is $\lambda_{Q,v}(P) = -\log \delta_v(P, Q)$. Changing these norms changes the function by a bounded amount, which is absorbed by the $O(1)$ term below. When the target is algebraic over k, one first fixes an extension of $|\cdot|_v$ to $\overline{k}$, or equivalently a compatible embedding $\iota_v : \overline{k} \hookrightarrow \overline{k_v}$, and applies the same formula to $\iota_v(Q)$. This choice is part of the local target data; no canonical embedding is being asserted.

Theorem 4.1 (simultaneous-place Roth theorem)**.** *Let k be a number field, let T be a finite set of places, and at each $v \in T$ choose one algebraic target $\alpha_v \in \mathbb{P}^1(\overline{k})$, with $\alpha_v = \infty$ allowed, and a compatible embedding $\iota_v : \overline{k} \hookrightarrow \overline{k_v}$. For every $\epsilon > 0$, outside a finite set of $t \in k$,*

$$\sum_{v \in T} n_v \lambda_{\iota_v(\alpha_v),v}(t) \leq (2 + \epsilon)[k : \mathbb{Q}]h(t) + O(1),$$

where t is regarded as the projective point $(t : 1)$ and $\lambda_{\iota_v(\alpha_v),v}$ is the local proximity function just defined.

Theorem 4.1 is the projective form of [2, Theorem 6.4.1], whose targets are written in an affine coordinate. To pass to the displayed form, choose $\beta \in k$ different from every finite target and put $\phi(z) = (z-\beta)^{-1}$. Then every $\phi(\alpha_v)$ is finite, including $\phi(\infty) = 0$. Apart from the single point $t = \beta$, which may be added to the exceptional set, functoriality of local Weil functions and heights under this fixed projective transformation gives

$$\lambda_{\alpha_v,v}(t) = \lambda_{\phi(\alpha_v),v}(\phi(t)) + O_v(1), \qquad h(\phi(t)) = h(t) + O_\phi(1).$$

Here the chosen embeddings into the local algebraic closures are retained when ϕ is applied. Applying the cited affine theorem to $\phi(t)$ and summing the finitely many bounded local errors proves the stated projective version, including a target at infinity. Section 5 applies this theorem to a rational function of degree $N-1$, producing the coefficient $2N-4$.

Theorem 4.2 (Northcott finiteness theorem)**.** *For fixed D and H, the set of algebraic numbers α with $[\mathbb{Q}(\alpha):\mathbb{Q}] \le D$ and $h(\alpha) \le H$ is finite.*

See [2, Theorem 1.6.8].

4.2. **Congruence expansion and counting.** We shall use congruence expansion in two forms. The random-walk estimate is used to choose a short perturbation of a prescribed group element. The geometric-ball estimate is used later to count trace classes. Both follow from strong approximation, superapproximation for bounded prime powers, the transfer of a spectral gap from Cayley graphs to surface covers, and uniform lattice-point counting.

Let y be an algebraic Fuchsian specialization, let $\Delta = \rho_y(\Pi^{(2)})$, and retain the smooth $\mathcal{O}_{k_y,S}$-model $\mathcal{H}$ of $\mathrm{SL}_1(B_y)$ from (17). The restriction of scalars $\mathbf{G} = \mathrm{Res}_{k_y/\mathbb{Q}}\,\mathrm{SL}_1(B_y)$ is an algebraic group over $\mathbb{Q}$. By contrast, the finite groups used for congruence counting are $\mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c})$. Lemma 2.5 verifies that the connected $\mathbb{Q}$-Zariski closure is the connected, semisimple, simply connected group $\mathbf{G}$. Congruence reduction is applied to the linear group $\Delta < \mathrm{SL}_2$. Its congruence kernels, transported through $\Delta \simeq \overline{\Delta}$, are finite-index subgroups of the projective surface lattice and therefore define covers of the fixed surface. The optional central element in the full inverse image affects indices by at most two and is absorbed in fixed constants.

Let q_0 be the product of the rational primes below S. Rational saturation gives $\mathcal{O}_{k_y,S} = \mathcal{O}_{k_y}[1/q_0]$. Form the smooth affine $\mathbb{Z}[1/q_0]$-group scheme

$$\mathcal{G}_{\mathrm{res}} = \mathrm{Res}_{\mathcal{O}_{k_y,S}/\mathbb{Z}[1/q_0]}\,\mathcal{H}.$$

Its generic fiber is $\mathbf{G}$, and for every integer m prime to q_0,

$$\mathcal{G}_{\mathrm{res}}(\mathbb{Z}[1/q_0]/m\mathbb{Z}[1/q_0]) = \mathcal{H}(\mathcal{O}_{k_y,S}/m\mathcal{O}_{k_y,S}).$$

Choose a faithful $\mathbb{Q}$-representation of $\mathbf{G}$. After enlarging S, spreading out gives a closed immersion

$$\iota : \mathcal{G}_{\mathrm{res}} \hookrightarrow \mathrm{GL}_{M,\mathbb{Z}[1/q_0]}$$

for one fixed integer M. Consequently $\iota(\Delta) \subset \mathrm{GL}_M(\mathbb{Z}[1/q_0])$, and reduction of this matrix group modulo m is exactly the reduction of Δ modulo $m\mathcal{O}_{k_y,S}$. We use this integral model to apply strong approximation and superapproximation to the same congruence quotients.

Theorem 4.3 (strong approximation in the form used here)**.** *After deleting finitely many rational primes, there is a constant C_{sa} such that the image $Q_{\mathfrak{c}}$ of Δ modulo every prime-to-S ideal $\mathfrak{c}$ satisfies*

$$[\mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c}) : Q_{\mathfrak{c}}] \le C_{\mathrm{sa}}. \tag{31}$$

Moreover,

$$\#Q_{\mathfrak{c}} \asymp_{y,S} (\mathrm{N}\,\mathfrak{c})^3, \tag{32}$$

uniformly in $\mathfrak{c}$.

We explain why strong approximation applies over the coefficient ring used in the congruence quotients. Put

$$A_{\mathrm{ad}} = \mathbb{Z}\big[\mathrm{tr}(\mathrm{Ad}\, h) : h \in \Delta\big] \subset k_y.$$

For $h \in \mathrm{SL}_2$,

$$\mathrm{tr}(\mathrm{Ad}\, h) = (\mathrm{tr}\, h)^2 - 1 = \mathrm{tr}(h^2) + 1.$$

The invariant-trace-field identity, together with its invariance under finite-index passage, therefore gives $\mathrm{Frac}(A_{\mathrm{ad}}) = k_y$. Because Δ is finitely generated, finite generation of matrix trace algebras [19, 12] makes A_{ad} a finitely generated trace ring. After inverting finitely many rational primes, it is the localization of an order in k_y. Enlarge the rationally saturated set S to contain every prime ideal dividing the conductor of that order. Localization then identifies the order with the full ring of S-integers, so

$$A_{\mathrm{ad}}[1/q_0] = \mathcal{O}_{k_y,S}.$$

Apply [25, Theorem 9.1.1]; see also [17, 18]. That theorem first supplies a localization of A_{ad} and a smooth simply connected semisimple model $\mathcal{H}_{\mathrm{W}}$ of the k_y-group $\mathrm{SL}_1(B_y)$, for which the closure of Δ in the product of the retained local integral groups is open. Enlarge S to contain the primes occurring in that localization. The preceding equality then permits base change of $\mathcal{H}_{\mathrm{W}}$ to $\mathcal{O}_{k_y,S}$. Its Weil restriction to $\mathbb{Z}[1/q_0]$ and $\mathcal{G}_{\mathrm{res}}$ both have generic fiber

$$\mathbf{G} = \mathrm{Res}_{k_y/\mathbb{Q}}\, \mathrm{SL}_1(B_y).$$

After one further localization, a generic-fiber isomorphism and its inverse extend to an isomorphism between these two models. Thus Weisfeiler's model, initially defined from the trace ring, agrees with the congruence model $\mathcal{H}$ outside the enlarged finite set S. Lemma 2.5 verifies that the connected $\mathbb{Q}$-Zariski closure is all of the simply connected semisimple group $\mathbf{G}$. Consequently the closure $\widehat{\Delta}$ of Δ is an open subgroup of

$$K = \prod_{p \nmid q_0} \mathcal{G}_{\mathrm{res}}(\mathbb{Z}_p) = \prod_{p \nmid q_0} \mathcal{H}(\mathcal{O}_{k_y} \otimes_{\mathbb{Z}} \mathbb{Z}_p).$$

Put $C_{\mathrm{sa}} = [K : \widehat{\Delta}]$. For every prime-to-$S$ ideal $\mathfrak{c}$, smoothness and the Chinese remainder theorem make the reduction map

$$r_{\mathfrak{c}} : K \longrightarrow \mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c})$$

surjective. Since the target is finite and Δ is dense in $\widehat{\Delta}$, $r_{\mathfrak{c}}(\widehat{\Delta}) = r_{\mathfrak{c}}(\Delta) = Q_{\mathfrak{c}}$. For a surjective homomorphism from a compact group to a finite group, the index of the image of an open subgroup is at most the index of that open subgroup. Hence

$$[\mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c}) : Q_{\mathfrak{c}}] \le [K : \widehat{\Delta}] = C_{\mathrm{sa}},$$

which proves (31) simultaneously for every prime-to-S ideal, with no restriction on its prime exponents.

It remains to calculate the size of the finite quotient. At a good prime $\mathfrak{p}$, with $q = \mathrm{N}\,\mathfrak{p}$, smooth lifting in relative dimension three and the fact that every quaternion algebra over a finite field splits give

$$\#\mathcal{H}(\mathcal{O}_{k_y}/\mathfrak{p}^e) = q^{3(e-1)} \#\,\mathrm{SL}_2(\mathbb{F}_q) = q^{3e}(1 - q^{-2}).$$

The Chinese remainder theorem and $\prod_{\mathfrak{p}}(1 - (\mathrm{N}\,\mathfrak{p})^{-2}) > 0$ show that $\#\mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c}) \asymp (\mathrm{N}\,\mathfrak{c})^3$. Equation (31) now proves (32). This use of strong approximation provides only the bounded index. The spectral gap required later comes from superapproximation.

Fix a finite symmetric generating set Ω of Δ. The *lazy random walk* used below starts at the identity and, independently at each step, stays in place with probability $1/2$ and otherwise right-multiplies by a uniformly chosen element of Ω. We denote its position after L steps by W_L.

Thus a walk with starting point g_0 ends at g_0W_L. Laziness makes the transition operator aperiodic and removes a possible spectral obstruction at the eigenvalue -1.

Under the closed immersion ι defined above, $\Omega \subset \mathrm{GL}_M(\mathbb{Z}[1/q_0])$, and Lemma 2.5 says that the identity component of its $\mathbb{Q}$-Zariski closure is $\iota(\mathbf{G})$, hence is perfect. Thus [20, Theorem 1] applies to the reductions of this matrix group modulo M_0^D, with M_0 square-free and prime to q_0. By the defining property of $\mathcal{G}_{\mathrm{res}}$, the resulting reduction image is

$$Q_{M_0^D\mathcal{O}_{k_y,S}}.$$

If $\mathfrak{c}$ has all prime-ideal exponents at most D, let M_0 be the product of the rational primes below $\mathfrak{c}$. All such primes are unramified because the ramified rational primes were put in S; consequently

$$M_0^D\mathcal{O}_{k_y,S} \subseteq \mathfrak{c}.$$

Reduction therefore induces a surjection $Q_{M_0^D\mathcal{O}_{k_y,S}} \twoheadrightarrow Q_{\mathfrak{c}}$. For the symmetric lazy walk used here, functions on $Q_{\mathfrak{c}}$ pull back along the quotient map to an invariant subspace of functions on $Q_{M_0^D\mathcal{O}_{k_y,S}}$. Hence every nonconstant eigenvalue of the quotient walk already occurs upstairs, so the same normalized spectral-gap lower bound holds for $Q_{\mathfrak{c}}$.

Theorem 4.4 (mixing modulo prime squares)**.** *Use the generating set and lazy random walk defined above. After removing finitely many rational primes, there are constants $B, C, \delta > 0$ such that, for every remaining rational prime p, every starting point g_0, and every subset $\mathcal{A}$ of the reduction image modulo p^2,*

$$\Pr(g_0W_L \bmod p^2 \in \mathcal{A}) \le C\frac{\#\mathcal{A}}{\#Q_{p^2}} + p^B e^{-\delta L}. \tag{33}$$

Here reduction modulo p^2 means reduction modulo $p^2\mathcal{O}_{k_y,S}$, and $Q_{p^2} = Q_{p^2\mathcal{O}_{k_y,S}}$ is the image of Δ in that quotient. The constants are independent of p, L, g_0.

The bounded-power superapproximation theorem just discussed, applied with exponent 2, supplies a uniform spectral gap for these congruence quotients. Making the walk lazy removes a possible eigenvalue near -1. Starting from a point mass, the resulting L^2-distance from the uniform distribution, with L^2 formed using unnormalized counting measure on the finite quotient, is $O(e^{-\delta L})$. Cauchy–Schwarz bounds its mass on $\mathcal{A}$ by

$$\sqrt{\#\mathcal{A}}\,O(e^{-\delta L}) \le \sqrt{\#Q_{p^2}}\,O(e^{-\delta L}) \le p^B e^{-\delta L};$$

the last inequality follows from the dimension-three local count above. This proves (33). Only the fixed exponent two is used here.

Theorem 4.5 (congruence counting for moduli with bounded prime exponents)**.** *Fix a faithful cocompact Fuchsian character x, an algebraic faithful Fuchsian character y, and an integer $D \ge 1$. Let*

$$\mathcal{B}_R^x = \{g \in \Pi^{(2)} : d_x(o, go) \le R\}.$$

There is a constant $\delta_D > 0$ with the following property. If $\mathfrak{c} \subset \mathcal{O}_{k_y,S}$ is prime to S and every prime exponent of $\mathfrak{c}$ is at most D, then, in the range $\log \mathrm{N}\,\mathfrak{c} = O(R)$ and uniformly in a trace class $u \bmod \mathfrak{c}$,

$$\#\{g \in \mathcal{B}_R^x : \mathrm{tr}_y\, g \equiv u \pmod{\mathfrak{c}}\} \le \frac{e^{R+o(R)}}{\mathrm{N}\,\mathfrak{c}} + (\mathrm{N}\,\mathfrak{c})^{2+o(1)}e^{(1-\delta_D)R}. \tag{34}$$

The error terms are uniform for fixed x, y, D. In particular, the second term is absorbed by the first if, for some fixed $\eta > 0$,

$$3\log \mathrm{N}\,\mathfrak{c} \le (\delta_D - \eta)R. \tag{35}$$

Consequently, if $\log \mathrm{N}\,\mathfrak{c}_m \le Am + O(1)$*, choose, after* D, A, δ_D *have been fixed, a constant*

$$C > \frac{3A}{\delta_D} \tag{36}$$

and put $R = Cm$*. Then*

$$\max_u \#\{g \in \mathcal{B}_R^x : \mathrm{tr}_y\, g \equiv u \pmod{\mathfrak{c}_m}\} \le \frac{e^{R+o(m)}}{\mathrm{N}\,\mathfrak{c}_m}. \tag{37}$$

In this statement a *trace class* is an element $u \in \mathcal{O}_{k_y,S}/\mathfrak{c}$, and $\mathrm{tr}_y\, g \equiv u \pmod{\mathfrak{c}}$ means that the trace of the integral matrix representing g maps to u under the quotient homomorphism. The displayed $o(R)$-terms are uniform in $\mathfrak{c}$ and u once x, y, D and a bound for $\log \mathrm{N}\,\mathfrak{c}/R$ have been fixed.

The integral-model argument preceding Theorem 4.4 gives a uniform spectral gap for the Cayley graphs of all the quotients occurring in this theorem. We next explain how this discrete spectral gap yields the stated count in a hyperbolic displacement ball. Let $\bar\rho_x : \Pi^{(2)} \to \mathrm{PSL}_2(\mathbb{R})$ be the projective Fuchsian representation with character x. Since $\rho_y : \Pi^{(2)} \to \Delta$ is an isomorphism onto its image, put

$$\begin{aligned} G_\infty &= \mathrm{PSL}_2(\mathbb{R}), & \Lambda_x &= \bar\rho_x(\Pi^{(2)}),\\ \pi_{\mathfrak{c}} &: \Delta \longrightarrow Q_{\mathfrak{c}}, & K_{\mathfrak{c}} &= \ker(\pi_{\mathfrak{c}}),\\ \bar\pi_{\mathfrak{c}} &= \pi_{\mathfrak{c}} \circ \rho_y : \Pi^{(2)} \longrightarrow Q_{\mathfrak{c}}, & & \\ \Pi_{\mathfrak{c}} &= \ker(\bar\pi_{\mathfrak{c}}), & \Lambda_{x,\mathfrak{c}} &= \bar\rho_x(\Pi_{\mathfrak{c}}). \end{aligned}$$

Let m_∞ be Haar measure on G_∞, and let $B_R \subset G_\infty$ be the set of elements moving o by at most R. Because $K_{\mathfrak{c}}$ is the kernel of a quotient map, its pullback $\Pi_{\mathfrak{c}}$ is normal in $\Pi^{(2)}$. Thus the associated surface cover is regular (Galois). Its deck group is $Q_{\mathfrak{c}}$, and its quotient graph for the fixed generating set is the Cayley graph whose spectral gap was obtained above. The Brooks–Burger transfer [3, 4], specifically Burger's [4, Théorème B], gives a number $\lambda_D > 0$, independent of $\mathfrak{c}$, such that

$$\lambda_1(\Lambda_{x,\mathfrak{c}}\backslash\mathbb{H}^2) \ge \lambda_D.$$

Let $p : \widetilde{G} = \mathrm{SL}_2(\mathbb{R}) \to G_\infty$ be the central covering, and put

$$\widetilde{\Lambda}_x = p^{-1}(\Lambda_x), \qquad \widetilde{\Lambda}_{x,\mathfrak{c}} = p^{-1}(\Lambda_{x,\mathfrak{c}}).$$

The natural maps

$$\widetilde{G}/\widetilde{\Lambda}_x \simeq G_\infty/\Lambda_x, \qquad \widetilde{G}/\widetilde{\Lambda}_{x,\mathfrak{c}} \simeq G_\infty/\Lambda_{x,\mathfrak{c}}$$

identify the corresponding unitary representations and the averages over a displacement ball with the averages over its full inverse image. Inversion, $g\widetilde{\Lambda} \mapsto \widetilde{\Lambda}g^{-1}$, identifies the right homogeneous spaces used here with the left-quotient convention for surface covers and in the cited lattice-counting results. Set

$$\lambda'_D = \min\{\lambda_D, 1/4\}, \qquad c_D = \frac{1 - \sqrt{1 - 4\lambda'_D}}{2} > 0.$$

Let $K_\infty = \mathrm{PSO}(2)$ be a maximal compact subgroup of G_∞, and put $\widetilde{K_\infty} = p^{-1}(K_\infty) = \mathrm{SO}(2)$. A nontempered irreducible constituent of $L^2_0(G_\infty/\Lambda_{x,\mathfrak{c}})$ is a complementary-series representation with parameter $s \in (1/2, 1)$, Laplace eigenvalue $s(1-s)$, and K_∞-finite matrix coefficients bounded, up to a polynomial factor, by $e^{-(1-s)t}$ on the Cartan ray. The displayed lower bound gives $1 - s \ge c_D$; see [13] for the complementary-series classification and [5] for the almost-L^2 matrix-coefficient criterion. Since Haar measure in Cartan coordinates has density $\asymp e^t\,dt$, choosing

$$p_D > \max\{2, c_D^{-1}\}$$

shows uniformly in $\mathfrak{c}$ that the representations on $L^2_0(\widetilde{G}/\widetilde{\Lambda}_{x,\mathfrak{c}})$ are L^{p_D+}. Here this notation means that every $\widetilde{K}_\infty$-finite matrix coefficient belongs to $L^{p_D+\varepsilon}(\widetilde{G})$ for every $\varepsilon > 0$.

The algebraic group SL_2 is simply connected and $\mathbb{R}$-isotropic, so its group of real points is within the scope of [8, Theorem 4.5]. For the normalized averages $\widetilde{\beta}_R$ on the inverse images of the displacement balls, that theorem gives, after choosing its auxiliary exponent sufficiently small,

$$\|\pi^0_{\mathfrak{c}}(\widetilde{\beta}_R)\| \le C_D\, m_\infty(B_R)^{-\kappa_D}$$

where $\pi^0_{\mathfrak{c}}$ denotes the averaging representation on the mean-zero subspace of $L^2(\widetilde{G}/\widetilde{\Lambda}_{x,\mathfrak{c}})$. Here $C_D, \kappa_D > 0$ are independent of $\mathfrak{c}$; the same estimate holds for the standard inner and outer thickenings of the balls. Transporting this operator-norm bound through the two quotient identifications above gives the same quantitative mean-ergodic estimate on G_∞, with constants independent of $\mathfrak{c}$.

Hyperbolic displacement balls are Hölder well rounded: thickening or shrinking the boundary by a small fixed neighborhood changes their Haar measure by a controlled power of the neighborhood size. Thus [8, Theorem 1.9 and Corollaries 1.11, 1.13] turn the preceding bound into a lattice-point estimate whose absolute error is $O_D(e^{(1-\delta_D)R})$ for some $\delta_D > 0$. The injectivity neighborhood may be chosen uniformly: each surface in the family covers the same fixed compact surface, so no new shorter closed geodesic is introduced. Moreover, a quotient element has a representative in the fixed base lattice Λ_x, and its fiber is the corresponding translate of $\Lambda_{x,\mathfrak{c}}$. The estimate is uniform over these translated cosets by [8, Remark 1.12]. Hence both the congruence subgroup and the chosen element of its finite quotient may vary in the count below.

With this normalization, uniform lattice counting gives, for every quotient element q,

$$\#\{g \in \mathcal{B}^x_R : \bar{\pi}_{\mathfrak{c}}(g) = q\} = \frac{m_\infty(B_R)}{m_\infty(G_\infty/\Lambda_x)\#Q_{\mathfrak{c}}} + O_D(e^{(1-\delta_D)R});$$

see [8, Theorem 1.9 and Corollaries 1.11, 1.13]. Here $m_\infty(B_R) = e^{R+O_x(1)}$. We also need the size of a trace fiber in a finite quotient. The following direct matrix count in $A = \mathcal{O}_{k_y}/\mathfrak{p}^e$ gives

$$\#\{g \in \mathrm{SL}_2(A) : \operatorname{tr} g = t\} \le (e+1)(\mathrm{N}\,\mathfrak{p})^{2e}.$$

To obtain it, write $g = \left(\begin{smallmatrix} a & b \\ c & t-a \end{smallmatrix}\right)$. The determinant condition is $bc = a(t-a) - 1$. For each of the $(\mathrm{N}\,\mathfrak{p})^e$ choices of a, partition b by its valuation between 0 and e; summing the corresponding numbers of solutions for c gives at most $(e+1)(\mathrm{N}\,\mathfrak{p})^e$ pairs (b,c). Write $\mathfrak{c} = \prod_{\mathfrak{p}} \mathfrak{p}^{e_{\mathfrak{p}}}$. At every retained good prime the chosen integral splitting identifies $\mathcal{H}(\mathcal{O}_{k_y,\mathfrak{p}}/\mathfrak{p}^e)$ with $\mathrm{SL}_2(\mathcal{O}_{k_y,\mathfrak{p}}/\mathfrak{p}^e)$, and reduced trace becomes ordinary matrix trace. Thus the preceding calculation and the Chinese remainder theorem give

$$\#\{h \in \mathcal{H}(\mathcal{O}_{k_y,S}/\mathfrak{c}) : \operatorname{tr} h = u\} \le (\mathrm{N}\,\mathfrak{c})^2 \prod_{\mathfrak{p}\mid\mathfrak{c}} (e_{\mathfrak{p}} + 1).$$

If $e_{\mathfrak{p}} \le D$, then

$$\prod_{\mathfrak{p}\mid\mathfrak{c}} (e_{\mathfrak{p}} + 1) \le (D+1)^{\omega(\mathfrak{c})} = (\mathrm{N}\,\mathfrak{c})^{o(1)}.$$

Here $\omega(\mathfrak{c})$ is the number of distinct prime-ideal divisors of $\mathfrak{c}$. The last estimate is uniform for fixed k_y, D: for every $\eta > 0$, the prime ideals with $\mathrm{N}\,\mathfrak{p} \le (D+1)^{1/\eta}$ contribute a fixed constant, while for every remaining prime ideal $D+1 \le (\mathrm{N}\,\mathfrak{p})^\eta$. Hence the product is at most $C_{k_y,D,\eta}(\mathrm{N}\,\mathfrak{c})^\eta$. Intersecting a trace fiber with $Q_{\mathfrak{c}}$ can only decrease its size, whereas Theorem 4.3 gives $\#Q_{\mathfrak{c}} \asymp (\mathrm{N}\,\mathfrak{c})^3$. Thus a trace fiber in $Q_{\mathfrak{c}}$ has size at most $(\mathrm{N}\,\mathfrak{c})^{2+o(1)}$ and relative density at most $(\mathrm{N}\,\mathfrak{c})^{-1+o(1)}$. Summing the quotient-element count over that fiber proves (34). The quotient of the error term by the main term is

$$(\mathrm{N}\,\mathfrak{c})^{3+o(1)} e^{-\delta_D R},$$

which tends to zero under the fixed positive margin in (35).

The constants are not claimed to be uniform as $D \to \infty$. In each application, D is fixed first and the radius constant is then chosen to satisfy the strict inequality (36).

4.3. **Algebraic geometry over finite fields.**

Theorem 4.6 (Lang–Weil consequence)**.** *Let $\mathcal{G}$ be a smooth affine group scheme over $\mathbb{Z}[1/M]$ whose geometric fibers are connected, and let $0 \neq \Phi \in \mathbb{Q}[\mathcal{G}_{\mathbb{Q}}]$. After enlarging M, so that every remaining fiber is geometrically integral, Φ extends to the model, and the zero locus of its reduction has dimension at most $\dim \mathcal{G}_{\mathbb{F}_p} - 1$, one has*

$$\frac{\#\{g \in \mathcal{G}(\mathbb{F}_p) : \Phi(g) = 0\}}{\#\mathcal{G}(\mathbb{F}_p)} = O_\Phi(p^{-1})$$

for every $p \nmid M$. The same bound, up to a fixed constant, holds in a subgroup of uniformly bounded index.

Indeed, if $d = \dim \mathcal{G}_{\mathbb{F}_p}$, Lang–Weil gives $\#\mathcal{G}(\mathbb{F}_p) = p^d + O(p^{d-1/2})$, uniformly after the fixed model has been chosen. Applied to the finitely many top-dimensional components of the zero locus, whose dimensions are at most $d-1$, it gives $O_\Phi(p^{d-1})$ points there. Division proves the displayed ratio. If $H_p < \mathcal{G}(\mathbb{F}_p)$ has index at most C, then

$$\frac{\#(H_p \cap V(\Phi))}{\#H_p} \leq C\,\frac{\#V(\Phi)(\mathbb{F}_p)}{\#\mathcal{G}(\mathbb{F}_p)},$$

which proves the bounded-index assertion. This is the only use of Lang–Weil: it permits one fixed prime of good reduction to be chosen so that a specified nonzero regular function is nonvanishing on more than three quarters of the congruence image; see [14].

4.4. **Hyperbolic counting and arithmeticity.** We use the following geometric facts for a fixed cocompact Fuchsian metric.

Orbit counting: For a finite-index subgroup of a closed surface group, $\#\mathcal{B}_R^x = e^{R+O_x(1)}$.

Metric comparison: Word length and displacement in the fixed metric are quasi-isometric.

Centralizers: The centralizer of a nonidentity element is infinite cyclic, and a cocompact torsion-free Fuchsian group has no nontrivial parabolic element.

To verify the orbit estimate, choose a compact fundamental domain of diameter D_0. The translates whose orbit points lie in a ball of radius $R - D_0$ are contained in the radius-R hyperbolic ball, while every translate meeting that ball has orbit point in the radius-$R + D_0$ ball. Division by the area of the fundamental domain and $\operatorname{area}(B_R) = 2\pi(\cosh R - 1)$ gives $\#\mathcal{B}_R^x = e^{R+O_x(1)}$; the same proof applies to a finite-index subgroup. The word/displacement comparison is the Švarc–Milnor lemma for the proper cocompact action on $\mathbb{H}^2$. Finally, every nonidentity element is hyperbolic: elliptics are excluded by torsion-freeness and parabolics by compactness of the quotient. A commuting element preserves its unique axis, and discreteness makes the group of translations along that axis cyclic. This proves the centralizer assertion.

Theorem 4.7 (finite-covolume corollary of Takeuchi's criterion)**.** *Let $\Lambda < \mathrm{PSL}_2(\mathbb{R})$ be a finite-covolume Fuchsian group, let $\widetilde{\Lambda}$ be its full inverse image in $\mathrm{SL}_2(\mathbb{R})$, and let $k = \mathbb{Q}(\operatorname{tr}\widetilde{\Lambda})$ be its ordinary trace field. Put*

$$K_2 = \mathbb{Q}((\operatorname{tr} g)^2 : g \in \widetilde{\Lambda}),$$

the invariant trace field. Suppose:

(i) *k is a number field;*

(ii) *every trace in* $\widetilde{\Lambda}$ *is an algebraic integer;*
(iii) *for every embedding* $\sigma : k \hookrightarrow \mathbb{C}$ *different from the distinguished real embedding and every* $g \in \widetilde{\Lambda}$*, one has* $\sigma(\operatorname{tr} g) \in [-2, 2]$.

Then Λ *is derived from a quaternion algebra and is arithmetic.*

This formulation is a corollary of Takeuchi's statements. Takeuchi denotes the ordinary trace field by k_1 and the invariant trace field by k_2. His Theorem 2, the criterion for being derived from a quaternion algebra used here, imposes the conjugate-trace condition for every nonidentity embedding of k_1. These are the embeddings covered by hypothesis (iii). The condition involving embeddings of k_1 whose restriction to k_2 is nonidentity occurs instead in Takeuchi's Theorem 1, his arithmetic criterion. Proposition 2 identifies the cocompact case with total reality and the interval $[-2, 2]$ [23, Theorem 2 and Proposition 2]. Theorem 2 now gives containment in the projective norm-one group associated with an order in the resulting admissible quaternion algebra. Both that norm-one group and Λ are lattices. Comparing covolumes shows that the containment has finite index; this is the asserted arithmeticity. The finite-covolume hypothesis is essential for this last step: an infinite-index subgroup can have the full boundary circle as its limit set without being a lattice. We use the corollary with $\Lambda = \Gamma^{(2)}$, after proving explicitly that its ordinary trace field is the invariant trace field of Γ. See also [16, Theorem 3.3.4 and Lemma 3.5.6].

5. A lower bound for the recurrence divisor

We first use Roth's approximation theorem to bound from below the norm of the product of two consecutive recurrence terms.

Lemma 5.1 (reduced numerator–denominator divisor)**.** *Let* k *be a number field, let* $f \in k(T)$ *have degree* $d \geq 2$*, and suppose every zero and pole of* f *on* $\mathbb{P}^1_k$ *is simple. Let* S *be a finite set of finite places containing the places at which a fixed integral model of* f*, or the reduced divisor of its zeros and poles, has bad reduction. For* $t \in k$ *which is neither a zero nor a pole, put*

$$\mathfrak{m}_{f,S}(t) = \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{|\operatorname{ord}_{\mathfrak{p}} f(t)|}.$$

Here $\operatorname{ord}_{\mathfrak{p}}$ *is the discrete valuation normalized by* $\operatorname{ord}_{\mathfrak{p}}(\mathfrak{p}) = 1$*. The absolute value in the exponent records both numerator and denominator primes, each with its full valuation. Here* reduced *means that common numerator and denominator factors have been cancelled; it does not mean that prime exponents have been truncated to one. For every* $\epsilon > 0$*, outside a finite set of* $t \in k$*,*

$$\log \mathrm{N}\, \mathfrak{m}_{f,S}(t) \geq (2d - 2 - \epsilon)[k : \mathbb{Q}]h(t) - O_{f,S,\epsilon}(1). \tag{38}$$

Proof. Let T_0 consist of S and all archimedean places. The product formula and functoriality of height give

$$\sum_{v \in M_k} n_v |\log |f(t)|_v| = 2[k : \mathbb{Q}]h(f(t)) = 2d[k : \mathbb{Q}]h(t) + O_f(1). \tag{39}$$

At a place $v \in T_0$, if $|f(t)|_v < 1$, then $-\log |f(t)|_v$, up to a bounded local term, is the proximity to a zero of f; if $|f(t)|_v > 1$, then $\log |f(t)|_v$ is similarly the proximity to a pole. Simplicity ensures that each target appears with multiplicity one. Pass to one fixed finite extension containing all zeros and poles. At each place over T_0, select a nearest zero or pole for the point under consideration. There are only finitely many possible selections, so Theorem 4.1 may be applied separately to each selection and the resulting inequality divided by the extension degree. Outside a finite exceptional set,

$$\sum_{v \in T_0} n_v |\log |f(t)|_v| \leq (2 + \epsilon)[k : \mathbb{Q}]h(t) + O_{f,S,\epsilon}(1). \tag{40}$$

The sum over the finite places $v \notin S$ in (39) is exactly $\log \mathrm{N}\, \mathfrak{m}_{f,S}(t)$. Subtract (40) from (39) and, if necessary, replace ϵ by a smaller positive number. □

Fix $N \geq 3$, and put

$$P_N(T) = q_{N-1}(T), \qquad Q_N(T) = q_N(T), \qquad F_N(T) = P_N(T)Q_N(T). \tag{41}$$

The rational function P_N/Q_N has degree $N-1$. Its finite zeros and poles are simple and disjoint, and infinity is one additional simple zero. Indeed, $q_j(T) = U_{j-1}(T/2)$ has distinct roots in characteristic zero, and (28) makes consecutive terms coprime.

Corollary 5.2 (two consecutive recurrence terms)**.** *Let k, S, N be fixed, with $N \geq 3$, and suppose that S contains the bad-reduction places of P_N/Q_N. For every $\epsilon > 0$, all but finitely many $t \in \mathcal{O}_{k,S}$ satisfy*

$$\log \mathrm{N}(F_N(t)\mathcal{O}_{k,S}) \geq (2N - 4 - \epsilon)\mathcal{J}_k(t) - O_{N,S,\epsilon}(1). \tag{42}$$

Proof. Cassini's identity shows that $P_N(t)$ and $Q_N(t)$ generate the unit ideal of $\mathcal{O}_{k,S}$. Therefore the reduced numerator–denominator ideal of $P_N(t)/Q_N(t)$ is precisely $F_N(t)\mathcal{O}_{k,S}$. Lemma 5.1, with $d = N-1$, gives the coefficient $2N - 4 - \epsilon$. Finally use Lemma 3.2; its bounded error is absorbed into the constant because k, N, S, ϵ are fixed. □

5.1. **Local density of square divisibility.** The selection argument in Lemma 5.5 requires the following density estimate modulo $\mathfrak{p}^2$.

Lemma 5.3 (square-divisor density)**.** *Fix $N \geq 3$. Outside a finite set of rational primes depending on N, k_y, and the integral model, the following holds. The excluded set contains the primes ramified in k_y, the bad and quaternion-ramified primes of the group model, the primes below $2N(N-1)$, the primes dividing the discriminant of F_N, and the exceptional primes in Theorem 4.3. If p is a remaining prime, then in the reduction image modulo p^2,*

$$\frac{\#\{g \in Q_{p^2} : F_N(\operatorname{tr} g) \equiv 0 \pmod{\mathfrak{p}^2} \text{ for some } \mathfrak{p} \mid p\}}{\#Q_{p^2}} \ll_{N,y,S} p^{-2}. \tag{43}$$

Proof. Put $A = \mathcal{O}_{k_y}/\mathfrak{p}^2$, where $q = \mathrm{N}\,\mathfrak{p}$. The stated exclusions make the reduction of F_N square-free. Hensel's lemma therefore gives

$$\#\{t \in A : F_N(t) = 0\} = O_N(1).$$

For one fixed $t \in A$, write a matrix of trace t as $\left(\begin{smallmatrix} u & b \\ c & t-u \end{smallmatrix}\right)$. Its determinant is one exactly when

$$bc = u(t-u) - 1.$$

There are q^2 choices for u. Splitting b according to its valuation 0, 1, or 2 gives at most $3q^2$ choices for (b, c) for each u. Thus a trace fiber contains at most $3q^4$ elements, whereas

$$\#\,\mathrm{SL}_2(A) = q^6(1 - q^{-2}).$$

It follows that the locus $F_N(\operatorname{tr} g) = 0 \pmod{\mathfrak{p}^2}$ has relative density $O_N(q^{-2})$. Cassini's identity ensures that the two factors of F_N have no common root. There are at most $[k_y : \mathbb{Q}]$ primes above p, and each has norm at least p. The union bound gives $O_{N,k_y}(p^{-2})$ in the full product of the local groups. Theorem 4.3 changes this by at most its fixed index, proving (43). □

Lemma 5.4 (height growth in matrix words)**.** *Let L be a number field, let S contain the archimedean places, and let $\Omega \subset \mathrm{GL}_d(\mathcal{O}_{L,S})$ be a fixed finite symmetric set. There is a constant C_Ω such that every entry of a word g of length at most n in Ω has height at most $C_\Omega n + C_\Omega$. If $P \in L[T]$ is fixed and S has been enlarged to contain the denominators of its coefficients, then, whenever $P(\operatorname{tr} g) \neq 0$,*

$$\log \mathrm{N}(P(\operatorname{tr} g)\mathcal{O}_{L,S}) \leq C_{\Omega,P}\, n + C_{\Omega,P}. \tag{44}$$

Proof. For a place v, put

$$\|M\|_v^+ = \max\{1, |M_{ij}|_v : 1 \le i, j \le d\}.$$

Matrix multiplication gives

$$\|AB\|_v^+ \le c_v \|A\|_v^+ \|B\|_v^+,$$

where $c_v = 1$ at a nonarchimedean place and $c_v = d$ at an archimedean place. Iterating this inequality for a word of length n, taking logarithms, multiplying by local degrees, and summing over all places shows that the projective height of its matrix entries is $O(n)$. Only finitely many places contribute because the generators and their inverses are fixed and S-integral. The height of each entry, and then of their sum $\operatorname{tr} g$, is therefore $O(n)$.

Applying the same local maximum estimate to the finitely many monomials of P gives the polynomial height inequality

$$h(P(\operatorname{tr} g)) \le (\deg P) h(\operatorname{tr} g) + O_P(1) = O_{\Omega,P}(n).$$

The value is S-integral. By the product formula, the logarithmic norm of its prime-to-S principal ideal is no larger than the sum of all negative local logarithms, which equals the sum of all positive local logarithms and is $[L : \mathbb{Q}]h(P(\operatorname{tr} g))$. This proves (44). □

Lemma 5.5 (selection of short perturbations with bounded valuations)**.** *Retain the hypotheses of Theorem 1.3. Put $\Delta = \rho_y(\Pi^{(2)})$, and fix a nonidentity $b \in \Pi^{(2)}$, an integer $N \ge 3$, and a sufficiently small $\theta > 0$. There are a rationally saturated finite set S of places of k_y, an integer $D \ge 1$, and, for all sufficiently large m, elements*

$$a_m = b^m h_m, \qquad |h_m| \le \theta m, \tag{45}$$

such that, with $s_m = \operatorname{tr}_y(a_m)$,

$$\mathcal{J}_{k_y}(s_m) \ge m\mathcal{J}_{k_y}(\operatorname{tr}_y b) - O_{b,y}(\theta m) - O_{b,y}(1), \tag{46}$$

$$d_x(o, a_m o) \le m\ell_x(b) + O_{b,x}(\theta m) + O_{b,x}(1), \tag{47}$$

and the nonzero ideal

$$\mathfrak{c}_m = F_N(s_m)\mathcal{O}_{k_y,S} \tag{48}$$

has every prime exponent at most D and satisfies

$$\log \mathrm{N}\, \mathfrak{c}_m = O_{N,b,y}(m). \tag{49}$$

There is a number $\theta_0 > 0$, depending on b, y, such that the constants implicit in (46), (47), and (49) may be chosen uniformly for $0 < \theta \le \theta_0$. The set S and the exponent D are allowed to depend on θ.

Proof. We specify the constants in the order needed for the final limiting argument. Enlarge a rationally saturated set S_0 so that the representation is integral, the restriction-of-scalars model has good reduction outside S_0, every rational prime ramified in k_y is removed, all primes below $2N(N-1)$ are removed, and the bad-reduction places of P_N/Q_N are included. Let h be the position after $L = \lfloor \theta m \rfloor$ steps of the lazy random walk defined before Theorem 4.4, and consider $a = b^m h$.

Choose one fixed finite Galois extension $E/\mathbb{Q}$, containing k_y, which splits the quaternion algebra and the characteristic polynomial of b. Only finitely many places w of E satisfy $\chi_w(\operatorname{tr}_y b) > 0$. At every such place choose one of the two global eigenvalues, denoted λ_w, so that $|\lambda_w|_w > 1$. The two eigenprojectors are

$$E_{w,+} = \frac{b - \lambda_w^{-1} I}{\lambda_w - \lambda_w^{-1}}, \qquad E_{w,-} = \frac{\lambda_w I - b}{\lambda_w - \lambda_w^{-1}}.$$

They satisfy

$$E_{w,+} + E_{w,-} = I, \qquad b^m = \lambda_w^m E_{w,+} + \lambda_w^{-m} E_{w,-}.$$

Define the fixed regular linear functions

$$L_{w,+}(h) = \operatorname{tr}(E_{w,+}h), \qquad L_{w,-}(h) = \operatorname{tr}(E_{w,-}h).$$

Since each projector has rank one, $L_{w,+}(I) = L_{w,-}(I) = 1$. Taking the trace after multiplying the projector decomposition by h gives

$$\operatorname{tr}_y(b^m h) = \lambda_w^m L_{w,+}(h) + \lambda_w^{-m} L_{w,-}(h). \tag{50}$$

For each w, take the product of all Galois conjugates over $\mathbb{Q}$ of the regular function $L_{w,+}$. Let Φ be the product of these finitely many functions, multiplied by one common nonzero denominator. Each factor is nonzero because it takes the value one at the identity. Consequently $\Phi \in \mathbb{Q}[\mathbf{G}] \setminus \{0\}$, and $\Phi(h) \neq 0$ implies $L_{w,+}(h) \neq 0$ for every expanding place w.

By Theorem 4.6, we may choose a sufficiently large prime r of good reduction for which the zero set of Φ occupies less than one eighth of the congruence image. Use the reduction modulo r associated with the model over the current ring $\mathcal{O}_{k_y,S_0}$. Mixing on this one fixed quotient shows, for all sufficiently large m, that

$$\Pr(\Phi(h) = 0) < \frac{1}{4}. \tag{51}$$

After obtaining this estimate, enlarge S_0 by all places above r. The enlargement removes the valuations above r from the divisor ideals but does not alter the fixed finite quotient in which the walk has already been evaluated. Suppose $\Phi(h) \neq 0$. Then every $L_{w,+}(h)$ is nonzero. Lemma 5.4 gives height $O(L)$ for the matrix entries of h. Since the coefficients of the projectors are fixed after b and the splitting field have been fixed, the height inequalities $h(\alpha\beta) \leq h(\alpha) + h(\beta)$ and $h(\alpha + \beta) \leq h(\alpha) + h(\beta) + O(1)$ give

$$h(L_{w,+}(h)),\ h(L_{w,-}(h)) = O_{b,y,E}(L). \tag{52}$$

For a nonzero algebraic number α, the product formula and $h(\alpha^{-1}) = h(\alpha)$ imply, at each fixed place w,

$$-\log|\alpha|_w \leq C_w h(\alpha), \qquad \log^+|\alpha|_w \leq C_w h(\alpha),$$

where the constant only reflects the fixed local-degree normalization. Applying this to (52) gives

$$-\log|L_{w,+}(h)|_w = O_{b,y,E}(L), \qquad \log^+|L_{w,-}(h)|_w = O_{b,y,E}(L). \tag{53}$$

Now $\log|\lambda_w|_w = \chi_w(\operatorname{tr}_y b) > 0$. Let c_0 be the minimum of these finitely many positive numbers, and let C_0 dominate the constants in (53). Choose $\theta > 0$ so small that $2C_0\theta < c_0$. This inequality defines an upper bound θ_0 for θ, and the constants remain uniform for $0 < \theta \leq \theta_0$. Since $L \leq \theta m$, the logarithm of the absolute value of the first term in (50) is at least

$$m\chi_w(\operatorname{tr}_y b) - C_0 L,$$

whereas that of the second is at most

$$-m\chi_w(\operatorname{tr}_y b) + C_0 L.$$

The first term therefore dominates exponentially. At a nonarchimedean place the ultrametric inequality makes the norm of the sum equal to the norm of the first term. At an archimedean place the ordinary triangle inequality loses only $O(1)$ in the logarithm. Thus

$$\log|\operatorname{tr}_y(b^m h)|_w \geq m\chi_w(\operatorname{tr}_y b) - O_{b,y,E}(L) - O_{b,y,E}(1).$$

At a nonarchimedean place with the displayed trace norm greater than one, $\chi_v(t) = \log|t|_v$; at an archimedean place the two quantities differ by at most $\log 2$, as in the proof of Lemma 3.2. Hence

$$\chi_w(\operatorname{tr}_y(b^m h)) \geq m\chi_w(\operatorname{tr}_y b) - O_{b,y,E}(L) - O_{b,y,E}(1).$$

Summing with the local-degree weights over the finitely many expanding places gives

$$\mathcal{J}_E(\operatorname{tr}_y(b^m h)) \geq m\mathcal{J}_E(\operatorname{tr}_y b) - O_{b,y,E}(\theta m) - O_{b,y,E}(1);$$

the remaining local summands on the left are nonnegative. Both traces lie in k_y, so the base-change formula (21) says

$$\mathcal{J}_E(\mathrm{tr}_y(b^m h)) = [E : k_y]\mathcal{J}_{k_y}(\mathrm{tr}_y(b^m h)), \quad \mathcal{J}_E(\mathrm{tr}_y\, b) = [E : k_y]\mathcal{J}_{k_y}(\mathrm{tr}_y\, b).$$

Division by $[E : k_y]$ proves (46) with the field normalization stated in the lemma.

For the displacement estimate, retain the basepoint o fixed earlier and choose a point z on the x-axis of b. The distance $d_x(o, z)$ is a fixed constant depending only on b, x, o. Then

$$d_x(o, b^m ho) \le d_x(o, z) + d_x(z, b^m z) + d_x(b^m z, b^m ho) \le m\ell_x(b) + 2d_x(o, z) + d_x(o, ho).$$

The displacement of a word is at most the sum of the displacements of its letters. Since the generating set is fixed and $|h| \le \theta m$, this proves (47).

The word length of $b^m h$ is $O(m)$. Apply Lemma 5.4 with $P = F_N$. After enlarging S_0 by the fixed coefficient denominators, it gives a constant A_N, uniform for $0 < \theta \le \theta_0$, such that, whenever the value is nonzero,

$$\log \mathrm{N}(F_N(\mathrm{tr}_y(b^m h))\mathcal{O}_{k_y,S_0}) \le A_N m + O_N(1). \tag{54}$$

Shrink θ_0, if necessary, so that the coefficient of m in the lower bound (46) is positive for $0 < \theta \le \theta_0$. The distinguished real place makes $\mathcal{J}_{k_y}(\mathrm{tr}_y\, b) > 0$, so (46) implies that the traces under consideration eventually leave every fixed finite set. In particular, F_N does not vanish for all sufficiently large m on the event $\Phi(h) \ne 0$.

Choose a large constant Z, larger than every exceptional rational prime in Lemma 5.3 and Theorem 4.4, and enlarge S_0 by every place above a rational prime at most Z; call the result S. By Lemma 5.3 and Theorem 4.4, for a good rational prime $p > Z$,

$$\Pr(\mathfrak{p}^2 \mid F_N(\mathrm{tr}_y(b^m h)) \text{ for some } \mathfrak{p} \mid p) \le C_N p^{-2} + p^B e^{-\delta L}. \tag{55}$$

Theorem 4.4 is uniform in the starting point, so it applies with starting point b^m.

If some exponent in $F_N(\mathrm{tr}_y(b^m h))\mathcal{O}_{k_y,S}$ is greater than an integer D, then (54) shows that the rational prime below it is at most

$$Y_m = \exp\left(\frac{A_N m + O_N(1)}{D+1}\right).$$

Indeed, $(D+1)\log \mathrm{N}\,\mathfrak{p} \le A_N m + O_N(1)$, and the rational prime below $\mathfrak{p}$ is no larger than $\mathrm{N}\,\mathfrak{p}$. The union bound in (55) gives

$$\begin{aligned}&\Pr(\Phi(h) \ne 0 \text{ and some prime-to-}S \text{ exponent exceeds } D)\\ &\quad \le C_N \sum_{p>Z} p^{-2} + \exp\left(\left(\frac{(B+1)A_N}{D+1} - \delta\theta\right) m + o(m)\right).\end{aligned} \tag{56}$$

Choose Z first so that the first term is less than $1/8$. Then choose the fixed integer D so that

$$D + 1 > \frac{(B+1)A_N}{\delta\theta}. \tag{57}$$

This makes the exponent in the second term negative. Here B, δ come from the fixed prime-square mixing estimate, A_N was fixed by the height bound after N, b, y were fixed, and $\theta > 0$ was fixed before D. Thus D is independent of m, although it may depend on N, b, y, θ, as allowed in the statement of the lemma. For large m, (56) is less than $1/4$. Together with (51), the total probability of the excluded events is less than one. Hence some walk position h_m satisfies all the stated properties. Equation (49) follows from (54). □

6. Controlling fibers of two trace functions

An edge in the trace graph is the pair $(\mathrm{tr}_y(AH), \mathrm{tr}_y H)$. To recover enough distinct edges from a large set of group elements, we must show that no such pair is assumed exponentially often. The required bound is ultimately a divisor estimate over a number field.

Lemma 6.1 (quadratic norm divisor estimate)**.** *Let L be a number field and S a finite set of places containing the archimedean places. Fix $A > 0$. Uniformly in $d, N \in \mathcal{O}_{L,S}$ with $dN \neq 0$ and $h(d), h(N) \leq Am$, the number of pairs $u, v \in \mathcal{O}_{L,S}$ satisfying*

$$h(u), h(v) \leq Am, \qquad u^2 - dv^2 = N$$

is at most $\exp(o_{A,L,S}(m))$.

Proof. Suppose first that d is not a square in L, and put

$$E = L(T), \qquad T^2 = d.$$

This is a quadratic field extension of L. Let S_E be the places of E above S, and let $\mathcal{O}_{E,S_E}$ be the integral closure of $\mathcal{O}_{L,S}$. Since T is integral over $\mathcal{O}_{L,S}$, every solution gives

$$\alpha = u + vT \in \mathcal{O}_{E,S_E}, \qquad \alpha\bar{\alpha} = N.$$

The map $(u, v) \mapsto \alpha$ is injective because $1, T$ is an L-basis of E.

We first count the possible prime-to-S_E principal ideals (α). For a prime $\mathfrak{P} \notin S_E$, integrality of both α and $\bar{\alpha}$ gives

$$0 \leq v_{\mathfrak{P}}(\alpha) \leq v_{\mathfrak{P}}(N).$$

If $\mathfrak{p}$ is the prime of L below $\mathfrak{P}$, then $v_{\mathfrak{P}}(N) \leq 2v_{\mathfrak{p}}(N)$, and at most two primes of E lie above $\mathfrak{p}$. The number of possible ideal patterns is therefore at most

$$\prod_{\substack{\mathfrak{p} \notin S \\ v_{\mathfrak{p}}(N) > 0}} \big(2v_{\mathfrak{p}}(N) + 1\big)^2. \tag{58}$$

The logarithmic norm of the prime-to-S numerator ideal of N is $O_{L,S,A}(m)$. Indeed, its valuation sum is the finite-place contribution to the height of N^{-1}, and $h(N^{-1}) = h(N) \leq Am$.

We claim that (58) is $\exp(o(m))$ uniformly in d, N. To see this, fix $T_0 > 1$. Only finitely many prime ideals of L have norm at most T_0, and their contribution to the logarithm of (58) is $O_{L,T_0}(\log m)$. For the remaining primes use $\log(2e + 1) \leq e \log 3$ and

$$\sum_{\mathrm{N}\,\mathfrak{p} > T_0} v_{\mathfrak{p}}(N) \leq \frac{1}{\log T_0} \sum_{\mathfrak{p}} v_{\mathfrak{p}}(N) \log \mathrm{N}\,\mathfrak{p} = O(m/\log T_0).$$

First let $m \to \infty$, and then $T_0 \to \infty$. This proves the required uniform subexponential estimate for the possible principal ideals.

It remains to count generators of each possible principal ideal. If α and α' arise from two admissible solutions with the same prime-to- S_E principal ideal, then

$$\varepsilon = \alpha/\alpha' \in \mathcal{O}_{E,S_E}^{\times}.$$

The height inequalities for sums, products, and quotients, together with $h(T) = \frac{1}{2}h(d)$, give

$$h(\alpha), h(\alpha') = O_{L,S,A}(m), \qquad h(\varepsilon) = O_{L,S,A}(m).$$

For an S_E-unit define its weighted logarithmic vector by

$$\ell_E(\varepsilon) = \big(n_w \log |\varepsilon|_w\big)_{w \in S_E},$$

where n_w is the usual local-degree weight. The product formula and the fact that $|\varepsilon|_w = 1$ outside S_E give the exact identity

$$\|\ell_E(\varepsilon)\|_1 = 2[E : \mathbb{Q}]h(\varepsilon). \tag{59}$$

Every such field E has degree at most $2[L:\mathbb{Q}]$, and S_E has cardinality bounded in terms of L, S. Kronecker's theorem says that a nonzero algebraic number has height zero if and only if it is a root of unity. Together with Northcott finiteness, it follows that there is a positive lower bound, depending only on $2[L:\mathbb{Q}]$, for the height of a non-torsion algebraic number of these degrees; the number of roots of unity is uniformly bounded as well [2, Sections 1.5–1.6]. By (59), distinct logarithmic vectors modulo torsion are therefore separated by a fixed positive distance. They lie in a ball of radius $O(m)$ in a space of uniformly bounded dimension. Packing gives

$$\#\{\varepsilon \in \mathcal{O}_{E,S_E}^{\times} : h(\varepsilon) = O(m)\} = m^{O_{L,S,A}(1)}.$$

Thus each possible principal ideal contributes only polynomially many solutions, and the nonsplit case follows.

It remains to treat the split case explicitly. Write $d = a^2$ with $a \in L^{\times}$. Since $d \in \mathcal{O}_{L,S}$, valuations outside S show that $a \in \mathcal{O}_{L,S}$, and $h(a) = \frac{1}{2}h(d)$. Every solution gives two nonzero S-integers

$$\alpha_+ = u + av, \qquad \alpha_- = u - av, \qquad \alpha_+\alpha_- = N.$$

The height bounds on u, v, a give $h(\alpha_+), h(\alpha_-) = O_{L,S,A}(m)$. The prime-to-$S$ ideal of α_+ divides that of N, so the same small-prime/large-prime argument gives $\exp(o(m))$ possible ideals. For each fixed ideal, two possible generators differ by an S-unit of height $O(m)$. The logarithmic-vector argument above, now in the fixed field L, gives only $m^{O_{L,S,A}(1)}$ generators. Finally, α_+ determines $\alpha_- = N/\alpha_+$, and then determines $u = (\alpha_+ + \alpha_-)/2$ and $v = (\alpha_+ - \alpha_-)/(2a)$. Hence the map from solutions to α_+ is injective. This proves the same subexponential bound in the split case and completes the proof. □

Lemma 6.2 (uniform fibers of two trace functions)**.** *Retain x, y, Δ from Theorem 1.3. Suppose that $A_m \in \Delta$ is noncentral and has word length at most Wm, where W is fixed. For every fixed $C > 0$, uniformly in $\xi, \zeta \in k_y$,*

$$\#\{H \in \Delta : d_x(o, Ho) \le Cm,\ \operatorname{tr}_y H = \xi,\ \operatorname{tr}_y(A_m H) = \zeta\} \le \exp(o_{C,W,x,y}(m)). \tag{60}$$

The same conclusion holds for squared traces, with at most four choices of sign.

Proof. Fix a splitting field of the quaternion algebra. Word length and x-displacement are comparable, so all matrix entries of A_m and of every H counted in (60) have logarithmic height $O_{C,W,x,y}(m)$ by Lemma 5.4, and they lie in one fixed ring of S-integers. If the fiber is empty, there is nothing to prove. If it is nonempty, choose one matrix H_0 in it. Then

$$\xi = \operatorname{tr}_y H_0, \qquad \zeta = \operatorname{tr}_y(A_m H_0)$$

also have height $O_{C,W,x,y}(m)$. Thus, whenever the fiber is nonempty, its prescribed trace values ξ, ζ satisfy the same height bound as its matrices. All constants below can therefore be chosen independently of ξ, ζ.

Write

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \qquad H = \begin{pmatrix} r & s \\ u & \xi - r \end{pmatrix}, \qquad w = \zeta - \delta\xi,$$

and put

$$\Delta_A = (\alpha + \delta)^2 - 4, \qquad y_0 = 2\beta r - (\alpha - \delta)s, \qquad \kappa = \beta\{\xi(\alpha - \delta) - 2w\}.$$

Since A is noncentral in a faithful cocompact Fuchsian group, $\operatorname{tr}(A)^2 \neq 4$: an element with trace squared equal to four is either central or projects to a parabolic element, and the latter cannot occur in a cocompact lattice. Hence $\Delta_A \neq 0$. Suppose first that $\beta \neq 0$. The determinant and

second trace equations are, respectively,

$$su = r(\xi - r) - 1, \tag{61}$$

$$\beta u = w - (\alpha - \delta)r - \gamma s. \tag{62}$$

Multiply (61) by β, substitute (62), and collect the terms in r. This gives

$$\beta r^2 - \{\beta\xi + (\alpha - \delta)s\}r + \beta + ws - \gamma s^2 = 0.$$

Multiplication by 4β and completion of the square gives

$$(y_0 - \beta\xi)^2 = \Delta_A s^2 + 2\kappa s + \beta^2(\xi^2 - 4),$$

because $\Delta_A = (\alpha - \delta)^2 + 4\beta\gamma$. Multiplying this identity by Δ_A and completing the square once more gives

$$(\Delta_A s + \kappa)^2 - \Delta_A(y_0 - \beta\xi)^2 = \kappa^2 - \Delta_A\beta^2(\xi^2 - 4). \tag{63}$$

The map

$$H \longmapsto (Y, V) = (y_0 - \beta\xi, \Delta_A s + \kappa)$$

is injective: V recovers s, then Y recovers r, and the second trace equation recovers u. All coefficients and solutions in (63) have height $O_{C,W,x,y}(m)$.

The right side of (63) equals

$$4\beta^2\big(\mathrm{tr}[A, H] - 2\big), \tag{64}$$

where $[A, H] = AHA^{-1}H^{-1}$. To verify (64), use

$$\mathrm{tr}[A, H] = \mathrm{tr}(A)^2 + \mathrm{tr}(H)^2 + \mathrm{tr}(AH)^2 - \mathrm{tr}(A)\,\mathrm{tr}(H)\,\mathrm{tr}(AH) - 2. \tag{65}$$

Indeed, substitution of $\mathrm{tr}(A) = \alpha + \delta$, $\mathrm{tr}(H) = \xi$, $\mathrm{tr}(AH) = \zeta$, and $w = \zeta - \delta\xi$ gives

$$\kappa^2 - \Delta_A\beta^2(\xi^2 - 4) = 4\beta^2\{\mathrm{tr}(A)^2 + \xi^2 + \zeta^2 - \mathrm{tr}(A)\xi\zeta - 4\},$$

which is (64) by (65).

If the right side is nonzero, put

$$d = \Delta_A, \qquad N_A = \kappa^2 - \Delta_A\beta^2(\xi^2 - 4), \qquad Y = y_0 - \beta\xi, \qquad V = \Delta_A s + \kappa.$$

Then (63) is exactly $V^2 - dY^2 = N_A$. The entry-height estimates above and the elementary height inequalities for sums and products give

$$h(d),\ h(N_A),\ h(Y),\ h(V) = O_{C,W,x,y}(m).$$

Thus all hypotheses of Lemma 6.1 hold over the fixed base splitting field and its fixed ring of S-integers; the lemma is uniform in the varying coefficient d and its associated quadratic algebra. It gives $\exp(o_{C,W,x,y}(m))$ possible pairs (Y, V), hence the same number of matrices H.

If $N_A = 0$, equivalently, if the right-hand side of (63) is zero, then $\mathrm{tr}[A, H] = 2$. Since $[A, H] \in \Delta$, its projective image is either the identity or a parabolic element. The latter cannot occur in a cocompact Fuchsian group, so $[A, H] = I$. The centralizer of a nonidentity surface-group element is cyclic. If its primitive hyperbolic generator has length $\ell_0 > 0$, the absolute trace of its n-th power is $2\cosh(|n|\ell_0/2)$. A prescribed signed trace therefore permits at most two exponents n, up to the fixed central choice of lift. Hence the degenerate fiber has $O(1)$ elements.

If $\beta = 0$ and $\gamma \neq 0$, replace A, H by their transposes. Transposition preserves determinants and each of the three traces $\mathrm{tr}\,A$, $\mathrm{tr}\,H$, and $\mathrm{tr}(AH)$, while the upper-right entry of A^{T} is $\gamma \neq 0$. The preceding argument therefore applies without changing the fiber. If $\beta = \gamma = 0$, the two trace equations determine both diagonal entries of H, while $\det H = 1$ fixes the product su. A nonzero product has $\exp(o(m))$ factorizations. Indeed, if $su = c \neq 0$, then (s) divides (c) outside S. The number of possible divisor ideals is bounded by

$$\prod_{\mathfrak{p} \notin S} (v_{\mathfrak{p}}(c) + 1) = \exp(o(m)),$$

by the small-prime/large-prime argument in (58), since $h(c) = O(m)$. For a fixed principal ideal, two possible generators differ by an S-unit; the bounded-degree height-separation and packing argument from Lemma 6.1 gives only $m^{O(1)}$ such generators of height $O(m)$. Once s is known, $u = c/s$ is fixed. Thus the number of factorizations is $\exp(o(m))$. If $su = 0$, the Fricke identity gives $\mathrm{tr}[A, H] = 2$, so the commutator is either the identity or a nontrivial parabolic element. The latter cannot occur in a cocompact Fuchsian group; hence $[A, H] = I$. Since A is noncentral diagonal with distinct eigenvalues, a matrix commuting with A is diagonal, and therefore $s = u = 0$. This leaves only the $O(1)$ diagonal solutions with the prescribed trace. Finally, two squared-trace conditions split into at most four signed conditions. □

7. Congruence counting in the trace graph

The following combinatorial lemma counts ordered length-two paths whose two edges share their middle vertex. It does not assume that the bipartite graph contains a Cartesian product.

Lemma 7.1 (congruence count for length-two paths)**.** *Let $\mathcal{R}$ be a Dedekind domain, let $P, Q \in \mathcal{R} \setminus \{0\}$ generate coprime ideals, and let $\varnothing \neq G \subset U \times V$, where $U, V \subset \mathcal{R}$ are finite. Put*

$$X_- = \{Qu - Pv : (u, v) \in G\}, \qquad X_+ = \{Qv - Pu : (u, v) \in G\},$$

and

$$M_{(PQ)}(V) = \max_{c \in \mathcal{R}/(PQ)} \#(V \cap (c + (PQ))).$$

Here (PQ) is the principal ideal generated by PQ, and $V \cap (c + (PQ))$ denotes the elements of V whose residue class in $\mathcal{R}/(PQ)$ is c. The nonemptiness of G implies that V is nonempty and $M_{(PQ)}(V) \geq 1$, so the denominators below are nonzero. Then

$$\#X_- \, \#X_+ \geq \frac{1}{M_{(PQ)}(V)} \sum_{v \in V} \deg_G(v)^2 \geq \frac{(\#G)^2}{\#V \, M_{(PQ)}(V)}. \tag{66}$$

Proof. An ordered length-two path is a triple (u, v, u') with $(u, v), (u', v) \in G$. Send it to

$$(Qu - Pv, \; Qv - Pu') \in X_- \times X_+.$$

For a fixed image (x, z), reduction of the two equations gives

$$v \equiv -P^{-1}x \pmod{(Q)}, \qquad v \equiv Q^{-1}z \pmod{(P)}.$$

The inverses exist because $(P) + (Q) = \mathcal{R}$. The Chinese remainder theorem therefore confines v to one residue class modulo (PQ). Once v, x, z are fixed, the two equations determine u, u' in the fraction field. Hence every fiber of this map has cardinality at most $M_{(PQ)}(V)$. There are $\sum_v \deg_G(v)^2$ paths. This proves the first inequality, and Cauchy–Schwarz gives $\sum_v \deg_G(v)^2 \geq (\#G)^2/\#V$. □

Theorem 7.2 (upper bound for the recurrence divisor)**.** *Retain x, y, Δ, assume the signed trace set at x satisfies the critical trace-growth bound, and fix $N \geq 3$. Let $A_m \in \Delta$ be noncentral, interpreted in $\Pi^{(2)}$ through ρ_y^{-1}, and suppose its word length is at most Wm for one fixed W. Put*

$$s_m = \mathrm{tr}_y(A_m), \qquad D_m = d_x(o, A_m o).$$

Suppose $D_m = O(m)$. Let S and D be fixed, with S rationally saturated, and assume that the nonzero ideals

$$\mathfrak{c}_m = \big(q_{N-1}(s_m) q_N(s_m)\big) \mathcal{O}_{k_y, S}$$

have every prime exponent at most D and satisfy

$$\log \mathrm{N}\, \mathfrak{c}_m \leq Am + O(1) \tag{67}$$

for one fixed constant A. Then

$$\log \mathrm{N}\,\mathfrak{c}_m \le \frac{2N-1}{2} D_m + o(m). \tag{68}$$

Proof. Let δ_D be the spectral exponent in Theorem 4.5. Choose

$$C > \frac{3A}{\delta_D} \tag{69}$$

and put $R = Cm$. Equations (67) and (69) give, for some fixed $\eta > 0$ and all sufficiently large m,

$$3 \log \mathrm{N}\,\mathfrak{c}_m \le (\delta_D - \eta) R.$$

Thus the absorption condition (35) holds. Theorem 4.5, in the form (37), gives

$$\max_c \#\{g \in \mathcal{B}_R^x : \mathrm{tr}_y\, g \equiv c \pmod{\mathfrak{c}_m}\} \le \frac{e^{R+o(m)}}{\mathrm{N}\,\mathfrak{c}_m}. \tag{70}$$

Here D, A, and δ_D are fixed before C is chosen; all four constants remain fixed as $m \to \infty$.

Form a simple bipartite graph whose left and right vertices are the trace values that occur in the first and second coordinates, respectively, and whose edge set is

$$\mathcal{G}_m = \{(u, v) = (\mathrm{tr}_y(A_m g), \mathrm{tr}_y\, g) : g \in \mathcal{B}_R^x\}. \tag{71}$$

Orbit counting gives $e^{R+O(1)}$ group elements in the ball, and Lemma 6.2 bounds every fiber of the two trace coordinates by $e^{o(m)}$. Therefore

$$\#\mathcal{G}_m \ge e^{R-o(m)}. \tag{72}$$

Let V_m be the second vertex class. Persistence of squared-trace collisions from x to y, together with the critical trace-growth bound at x, gives

$$\#V_m \le e^{R/2+o(m)}. \tag{73}$$

Indeed, the ball has at most $e^{R/2+o(R)}$ squared-trace values at x. Collision persistence from Lemma 2.2 allows no more squared-trace values at y, and each squared trace has at most two signed square roots. Here $R = Cm$, with C fixed before $m \to \infty$.

Put

$$V_m' = \left\{ v \in V_m : \deg_{\mathcal{G}_m}(v) \ge \frac{\#\mathcal{G}_m}{4\#V_m} \right\}, \qquad \mathcal{G}_m' = \{(u, v) \in \mathcal{G}_m : v \in V_m'\}.$$

The vertices outside V_m' are incident to fewer than $\#V_m \cdot \#\mathcal{G}_m/(4\#V_m) = \#\mathcal{G}_m/4$ edges, so at most one quarter of the edges is deleted. For every $(u, v) \in \mathcal{G}_m'$, choose one element $g_{u,v} \in \mathcal{B}_R^x$ satisfying

$$\mathrm{tr}_y(A_m g_{u,v}) = u, \qquad \mathrm{tr}_y(g_{u,v}) = v.$$

For each fixed $v \in V_m'$, the representatives chosen for its incident edges are distinct: one group element determines only one ordered pair of traces. The definition of V_m', together with (72) and (73), therefore gives at least $e^{R/2-o(m)}$ distinct chosen group elements with second trace v. Combining this lower bound with (70) gives

$$M_{\mathfrak{c}_m}(V_m') \le \frac{e^{R/2+o(m)}}{\mathrm{N}\,\mathfrak{c}_m}. \tag{74}$$

Indeed, if one residue class contains r values from V_m', the chosen representatives of their incident edges give at least $re^{R/2-o(m)}$ elements of $\mathcal{B}_R^x$ whose traces lie in that class. Equation (70) bounds their number by $e^{R+o(m)}/\mathrm{N}\,\mathfrak{c}_m$, which yields the displayed inequality. The passage to V_m' reconciles two different counts: congruence counting counts group elements, with all trace multiplicities retained, whereas Lemma 7.1 counts distinct trace values.

Apply Lemma 7.1 over $\mathcal{O}_{k_y,S}$, with

$$P = q_{N-1}(s_m), \qquad Q = q_N(s_m).$$

Cassini's identity makes (P) and (Q) coprime, and $\mathfrak{c}_m = (PQ)$. Thus $M_{\mathfrak{c}_m}(V_m') = M_{(PQ)}(V_m')$ in the notation of Lemma 7.1. Equations (72)–(74) give

$$\#X_- \, \#X_+ \geq e^{R-o(m)} \, \mathrm{N}\, \mathfrak{c}_m. \tag{75}$$

The recurrence identities show that every element of X_- and X_+ is a trace of a group element:

$$X_- = \{\mathrm{tr}_y(A_m^N g_{u,v}) : (u,v) \in \mathcal{G}_m'\}, \qquad X_+ = \{\mathrm{tr}_y(A_m^{-(N-1)} g_{u,v}) : (u,v) \in \mathcal{G}_m'\}.$$

For $(u,v) \in \mathcal{G}_m'$ and $j \geq 0$,

$$d_x(o, A_m^j g_{u,v} o) \leq R + jD_m,$$

and the analogous inequality holds for negative powers. For each of these two displacement balls, persistence of squared-trace equalities from x to y bounds the number of traces at y by the number of squared traces at x, up to a factor two for the choice of sign. The critical trace-growth bound therefore implies

$$\#X_- \, \#X_+ \leq \exp\left(R + \frac{2N-1}{2} D_m + o(m)\right). \tag{76}$$

The error in the estimate for X_- is $o(R + ND_m)$, and that for X_+ is $o(R + (N-1)D_m)$. Both are $o(m)$, because C and N are fixed at this stage and $D_m = O(m)$. Comparison of (75) and (76) proves (68). □

Remark 7.3 (sharpness of the trace-growth exponent for this argument). Suppose more generally that $N_\Gamma(X) \leq X^{\sigma+o(1)}$, and put $\alpha = \sigma/2$. A ball of radius R then has at most $\exp(\alpha R + o(R))$ trace values. In the preceding proof this gives

$$\#V_m \leq e^{\alpha R + o(m)}, \qquad \deg_{\mathcal{G}_m'}(v) \geq e^{(1-\alpha)R - o(m)}, \qquad M_{\mathfrak{c}_m}(V_m') \leq \frac{e^{\alpha R + o(m)}}{\mathrm{N}\, \mathfrak{c}_m}.$$

The two shifted trace supports are bounded at radii $R + ND_m$ and $R + (N-1)D_m$. Substitution in the length-two-path lower bound and the shifted-trace upper bound consequently gives

$$\log \mathrm{N}\, \mathfrak{c}_m \leq (2\sigma - 2)R + \frac{\sigma(2N-1)}{2} D_m + o(m).$$

At $\sigma = 1$, the additional term involving $R = Cm$ cancels and the coefficient used in Theorem 1.3 is recovered. A fixed power loss $\sigma > 1$ leaves a positive multiple of the auxiliary radius and does not prove the local theorem. The hypothesis $N_\Gamma(X) \leq X^{1+o(1)}$ is the critical case treated by the theorem.

8. Proof of the local expansion theorem

Proof of Theorem 1.3. For the identity, $\mathrm{tr}_y\, b = 2$, every local expansion exponent vanishes, and $\ell_x(b) = 0$; hence the assertion holds. Let $b \in \Pi^{(2)}$ be nonidentity. Its image under every faithful cocompact Fuchsian representation is hyperbolic in PSL_2, and hence every SL_2-lift is noncentral.

Fix, in this order, an integer $N \geq 3$, a number $0 < \epsilon < 2N - 4$, and a sufficiently small positive number θ; then fix the set S, exponent D, and sequence a_m given by Lemma 5.5. Finally choose the ball constant C required in the proof of Theorem 7.2. These data remain fixed as $m \to \infty$.

Equation (46) shows that $s_m = \mathrm{tr}_y(a_m)$ leaves every finite set. Corollary 5.2 therefore applies for all sufficiently large m. Combining it with Theorem 7.2 gives

$$\begin{aligned}(2N - 4 - \epsilon)&(m\mathcal{J}_{k_y}(\mathrm{tr}_y\, b) - O(\theta m) - O(1)) \\ &\leq \frac{2N-1}{2}(m\ell_x(b) + O(\theta m) + O(1)) + o(m).\end{aligned}$$

Divide first by m and let $m \to \infty$, with $N, \epsilon, \theta, S, D, C$ fixed. To take the next limit, repeat the entire construction for a family of positive values $\theta \to 0$. Each new value of θ is allowed to produce a new set S_θ, a new exponent bound D_θ, a new spectral exponent, and a new ball constant C_θ. For each value of θ, the limit in m is taken before θ is changed. Consequently these auxiliary data need not be uniform as $\theta \to 0$. The error coefficients in (46) and (47) are uniform for $0 < \theta \le \theta_0$, so this family of inequalities permits $\theta \to 0$. We then let $\epsilon \to 0$, and finally $N \to \infty$. Since

$$\lim_{N\to\infty} \frac{(2N-1)/2}{2N-4} = \frac{1}{2},$$

we obtain

$$\mathcal{J}_{k_y}(\operatorname{tr}_y b) \le \frac{\ell_x(b)}{2}.$$

This is (6). □

9. Arithmeticity and exclusion of transcendental characters

Proof of Theorem 1.1. Let x be the marked Fuchsian character of Γ.

Suppose first that x is algebraic. Apply Theorem 1.3 with $y = x$, and put $\Lambda = \Gamma^{(2)}$. Since Π is finitely generated and $\Pi/\Pi^{(2)}$ is an elementary abelian two-group, Λ has finite index in Γ. In particular, Λ is itself a torsion-free cocompact lattice. Choose an SL_2-lift ρ_x of the marked representation and set $\Delta_x = \rho_x(\Pi^{(2)})$; this is a linear lift of Λ. Its full inverse image is $\widetilde{\Lambda} = \{\pm h : h \in \Delta_x\}$, so the full inverse image and Δ_x generate the same ordinary trace field. The invariant-trace-field identity of Lemma 2.1 gives

$$\mathbb{Q}(\operatorname{tr}\Delta_x) = \mathbb{Q}(\operatorname{tr}(\widetilde{\gamma})^2 : \widetilde{\gamma} \in \widetilde{\Gamma}) = k_\Gamma = k_x.$$

Thus the ordinary trace field of Λ, computed from its full inverse image, is the invariant trace field of Γ. For every nonidentity $b \in \Pi^{(2)}$, the distinguished real place contributes $\ell_x(b)/2$ to $\mathcal{J}_{k_x}(\operatorname{tr}_x b)$ by (23), while every local summand is nonnegative. The upper bound (6) therefore forces equality and makes every other local summand zero. At every finite place, $\chi_v(\operatorname{tr}_x b) = 0$. If $\lambda + \lambda^{-1} = \operatorname{tr}_x b$, equation (22) gives $\max\{|\lambda|_v, |\lambda|_v^{-1}\} = 1$, so $|\lambda|_v = 1$. The ultrametric inequality yields $|\operatorname{tr}_x b|_v = |\lambda + \lambda^{-1}|_v \le 1$. Thus all traces of the chosen lift Δ_x are algebraic integers. The full inverse image $\widetilde{\Lambda}$ consists of the two lifts $\pm\rho_x(b)$, with $b \in \Pi^{(2)}$. Its traces are consequently $\pm\operatorname{tr}_x b$; algebraic integrality is unchanged by sign, and the central traces ± 2 satisfy the same conclusion.

At every nondistinguished archimedean place, the roots in (18) have modulus one. Thus

$$\sigma(\operatorname{tr}_x b)^2 - 2 = z + z^{-1} \in [-2, 2],$$

so $\sigma(\operatorname{tr}_x b)^2 \in [0, 4]$. It follows that $\sigma(\operatorname{tr}_x b)$ is real and belongs to $[-2, 2]$. The same interval contains its negative, so this verifies the conjugate-trace condition for every element of the full inverse image $\widetilde{\Lambda}$. These traces generate k_x; hence every embedding of k_x has real image and k_x is totally real. All hypotheses of Theorem 4.7 are now verified for the finite-covolume Fuchsian group Λ. Takeuchi's theorem therefore implies that Λ is arithmetic. Arithmeticity is invariant under commensurability, so Γ is arithmetic.

Suppose now that x is transcendental. Then $\dim Z_x > 0$, and Proposition 2.3 gives algebraic Fuchsian characters $y_j \in Z_x$ converging to x, all in the same Fuchsian component as x. Choose one of them. Since x is transcendental, $y_j \ne x$, whereas Corollary 1.4 gives $y_j = x$. This contradiction excludes the transcendental case, and the algebraic argument proves the theorem. □

Proof of Corollary 1.2. First suppose that Γ is torsion-free. The critical-growth assertion is Theorem 1.1. Linear growth implies the critical bound. If the signed trace set has bounded clustering, cover $[-X, X]$ by $O(X)$ intervals of radius one. Each interval contains a uniformly bounded number of trace values, so (3) holds.

For a general cocompact Fuchsian lattice, Selberg's lemma supplies a torsion-free finite-index subgroup $\Gamma_0 < \Gamma$ [1]. Its signed trace set is a subset of that of Γ, so each of the three hypotheses under consideration passes to Γ_0. The torsion-free case makes Γ_0 arithmetic. Arithmeticity is a commensurability property, and therefore Γ is arithmetic as well. □

10. Dependencies among the estimates

The following diagram records how the estimates are used in the proof. Each arrow denotes a logical dependence.

$$\text{simultaneous-place Roth theorem} \longrightarrow \text{Corollary } 5.2,$$

$$\text{prime-square mixing and Lang–Weil} \longrightarrow \text{Lemma } 5.5,$$

$$\text{number-field divisor count} \longrightarrow \text{Lemma } 6.2,$$

$$\text{bounded-exponent congruence counting} + \text{Lemma } 7.1 + \text{Lemma } 6.2 \longrightarrow \text{Theorem } 7.2,$$

$$\text{Corollary } 5.2 + \text{Lemma } 5.5 + \text{Theorem } 7.2 \longrightarrow \text{Theorem } 1.3,$$

$$\text{algebraic specialization} + \text{Theorem } 1.3 + \text{Thurston's asymmetric metric} \longrightarrow x \text{ is algebraic},$$

$$\text{Theorem } 1.3 \text{ at } x + \text{local nonnegativity} \longrightarrow \text{trace integrality and bounded conjugates},$$

$$\begin{array}{c}\text{trace integrality and bounded conjugates+Takeuchi's criterion}\\ \text{+commensurability invariance of arithmeticity}\end{array} \longrightarrow \text{Theorem } 1.1,$$

$$\text{Selberg's lemma} + \text{commensurability} \longrightarrow \text{Corollary } 1.2 \text{ for orbifolds}.$$

The first five lines establish the local expansion inequality. Algebraic specialization then shows that the original character is algebraic, and Takeuchi's criterion converts the local equality at that character into arithmeticity.

Acknowledgements

We thank Igor Rivin for helpful comments and a careful critical reading.

References

[1] R. C. Alperin, *An elementary account of Selberg's lemma*, Enseign. Math. (2), 33 (1987), pp. 269–273. e-periodica.

[2] E. Bombieri and W. Gubler, *Heights in Diophantine Geometry*, vol. 4 of New Mathematical Monographs, Cambridge University Press, Cambridge, 2006. doi:10.1017/CBO9780511542879.

[3] R. Brooks, *The spectral geometry of a tower of coverings*, J. Differential Geom., 23 (1986), pp. 97–107. doi:10.4310/jdg/1214439903.

[4] M. Burger, *Spectre du Laplacien, graphes et topologie de Fell*, Comment. Math. Helv., 63 (1988), pp. 226–252. doi:10.1007/BF02566764.

[5] M. Cowling, U. Haagerup, and R. Howe, *Almost L^2 matrix coefficients*, J. Reine Angew. Math., 387 (1988), pp. 97–110. doi:10.1515/crll.1988.387.97.

[6] S. Geninska and E. Leuzinger, *A geometric characterization of arithmetic Fuchsian groups*, Duke Math. J., 142 (2008), pp. 111–125. doi:10.1215/00127094-2008-002; arXiv:math/0609477.

[7] W. M. Goldman, *Topological components of spaces of representations*, Invent. Math., 93 (1988), pp. 557–607. doi:10.1007/BF01410200.

[8] A. Gorodnik and A. Nevo, *Counting lattice points*, J. Reine Angew. Math., 663 (2012), pp. 127–176. doi:10.1515/CRELLE.2011.096; arXiv:0903.1515.

[9] Y. Hao, *On trace sets of hyperbolic surfaces and a conjecture of Sarnak and Schmutz*, Groups Geom. Dyn., (2025). Published online first; doi:10.4171/GGD/932; arXiv:2410.05223.

[10] ———, *On the length sets of closed hyperbolic surfaces*, 2026. arXiv:2607.20808 [math.GT].

[11] M. Heusener and J. Porti, *The variety of characters in* $\mathrm{PSL}_2(\mathbb{C})$, Bol. Soc. Mat. Mexicana (3), 10 (2004), pp. 221–237. Special issue; arXiv:math/0302075.

[12] R. D. Horowitz, *Characters of free groups represented in the two-dimensional special linear group*, Comm. Pure Appl. Math., 25 (1972), pp. 635–649. doi:10.1002/cpa.3160250602.
[13] S. Lang, $\mathrm{SL}_2(\mathbb{R})$, vol. 105 of Graduate Texts in Mathematics, Springer-Verlag, New York, 1985. doi:10.1007/978-1-4612-5142-2.
[14] S. Lang and A. Weil, *Number of points of varieties in finite fields*, Amer. J. Math., 76 (1954), pp. 819–827. doi:10.2307/2372655.
[15] W. Luo and P. Sarnak, *Number variance for arithmetic hyperbolic surfaces*, Comm. Math. Phys., 161 (1994), pp. 419–432. doi:10.1007/BF02099785.
[16] C. Maclachlan and A. W. Reid, *The Arithmetic of Hyperbolic* 3*-Manifolds*, vol. 219 of Graduate Texts in Mathematics, Springer-Verlag, New York, 2003. doi:10.1007/978-1-4757-6720-9.
[17] C. R. Matthews, L. N. Vaserstein, and B. Weisfeiler, *Congruence properties of Zariski-dense subgroups. I*, Proc. London Math. Soc. (3), 48 (1984), pp. 514–532. doi:10.1112/plms/s3-48.3.514.
[18] M. V. Nori, *On subgroups of* $\mathrm{GL}_n(\mathbb{F}_p)$, Invent. Math., 88 (1987), pp. 257–275. doi:10.1007/BF01388909.
[19] C. Procesi, *The invariant theory of* $n \times n$ *matrices*, Adv. Math., 19 (1976), pp. 306–381.
[20] A. Salehi Golsefidy, *Super-approximation, II: the p-adic case and the case of bounded powers of square-free integers*, J. Eur. Math. Soc., 21 (2019), pp. 2163–2232. doi:10.4171/JEMS/883; arXiv:1602.00409.
[21] P. Sarnak, *Arithmetic quantum chaos*, in The Schur Lectures (1992), vol. 8 of Israel Math. Conf. Proc., Bar-Ilan University, Ramat Gan, 1995, pp. 183–236. Author's 1993 manuscript.
[22] P. Schmutz, *Arithmetic groups and the length spectrum of Riemann surfaces*, Duke Math. J., 84 (1996), pp. 199–215. doi:10.1215/S0012-7094-96-08407-0.
[23] K. Takeuchi, *A characterization of arithmetic Fuchsian groups*, J. Math. Soc. Japan, 27 (1975), pp. 600–612. doi:10.2969/jmsj/02740600.
[24] W. P. Thurston, *Minimal stretch maps between hyperbolic surfaces*, 1986. 1986 preprint; arXiv:math/9801039.
[25] B. Weisfeiler, *Strong approximation for Zariski-dense subgroups of semi-simple algebraic groups*, Ann. of Math. (2), 120 (1984), pp. 271–315. doi:10.2307/2006943.